

IMPERIAL

HRAC: HISTORY-RESIDUAL ADAPTIVE CLIPPING FOR BYZANTINE-ROBUST FEDERATED LEARNING

Author

Y. HUANG

CID: 02196656

Supervised by

DR STEFAN VLASKI

TBC

A Thesis submitted in fulfillment of requirements for the degree of
Master of Engineering in Electrical and Electronic Engineering

Department of Electrical and Electronic Engineering
Imperial College London
2026

Abstract

Server-side aggregation is a central vulnerability in federated learning. Existing Byzantine-robust aggregation is usually designed for worst-case Byzantine attacks, where malicious clients may send arbitrary updates to the server. However, such defences can be overly conservative under non-IID client data and protocol-compliant attacks such as label poisoning. In these settings, benign clients may naturally deviate from the majority behaviour, while mean aggregation, although less conservative, remains vulnerable to stronger model-poisoning attacks. This thesis proposes *HRAC* (History-Residual Adaptive Clipping), a server-side aggregation method that balances robustness and heterogeneity tolerance. HRAC combines an EMA-smoothed median norm cap, history-residual clipping, and delayed soft weighting based on residual-change statistics. Rather than filtering clients by their deviation from the population, HRAC compares each client with its own historical behaviour and limits the influence of unstable update dynamics. Fixed-run experiments on CIFAR-10 under IID and non-IID partitions show that robustness is attack-dependent: existing robust baselines can be conservative in heterogeneous settings, while mean aggregation remains insufficient against stronger model-poisoning attacks. HRAC provides a more balanced result in these runs, with its strongest improvements under MSA and under non-IID ALIE/IPM.

Declaration of Originality

I hereby declare that, unless otherwise indicated, the work presented in this thesis is entirely my own. To the best of my knowledge, it is original, and any ideas developed in collaboration with others have been appropriately acknowledged and referenced. I acknowledge the use of ChatGPT-5 (OpenAI, <https://chatgpt.com>) as a support tool for language editing and for checking the clarity of explanations. I confirm that the research design, implementation, experiments, analysis, interpretation, and conclusions are my own work, except where otherwise cited. All text included in this dissertation has been reviewed and edited by me, and I take full responsibility for its content.

Copyright Declaration

Attribution-NoDerivatives Licence (CC BY-ND 4.0)

The copyright of this thesis rests with the author. Unless otherwise indicated, its contents are licensed under a Creative Commons Attribution NoDerivatives 4.0 International Licence (CC BY-ND).

Under this licence, you may copy and redistribute the material in any medium or format for both commercial and non-commercial purposes. This on the condition that; you credit the author and do not distribute modified versions of the work.

When reusing or sharing this work, ensure you make the licence terms clear to others by naming the licence and linking to the licence text.

Please seek permission from the copyright holder for uses of this work that are not included in this licence or permitted under UK Copyright Law.

Contents

Abstract	i
Declaration of Originality	iii
Copyright Declaration	v
1 Introduction	1
1.1 Motivation	1
1.2 Project Aims and Contributions	2
2 Background and Related Work	3
2.1 Federated Learning and Data Heterogeneity	3
2.2 Byzantine-Robust Aggregation	4
2.3 Poisoning Attacks and Attack-Dependent Robustness	6
2.4 History-Aware and Lightweight Defences	7
2.5 Research Gap	8
3 Method and Implementation	9
3.1 Design Rationale	9
3.2 Problem Setup and Threat Model	10
3.3 Aggregation Pipeline	11
3.4 Core Components	11
3.4.1 Global Median Norm Cap	11
3.4.2 Per-Client History-Residual Clipping	13
3.4.3 Separated Aggregation and Statistics Paths	13
3.4.4 EMA History Updates	14
3.4.5 Soft Weighting from the ν Statistic	14
3.5 Software Integration and Logging	15
3.6 Hyperparameters	16
3.7 Convergence Statement	16

4 Experiments and Results	19
4.1 Experimental Setup	19
4.2 Experimental Results	20
4.3 Additional Analyses	23
4.3.1 Final and Best Accuracy Summary	23
4.3.2 Component Ablation Study	27
4.3.3 Diagnostic Analysis	28
4.4 Discussion and Limitations	30
Conclusions and future directions	35
A Convergence Analysis of HRAC	37
A.1 Introduction	37
A.2 Problem Setup	38
A.3 HRAC Aggregation Rule	39
A.3.1 Global norm clipping	39
A.3.2 Per-client history-residual clipping	40
A.3.3 Adaptive weighting	41
A.4 Assumptions	41
A.5 Basic Properties of HRAC	46
A.6 Aggregation Error Analysis	49
A.7 From Honest Conditional Centre to the True Gradient	51
A.8 One-Step Descent Analysis	51
A.9 Main Convergence Theorem	52
A.10 Weight Control from a d -Gap Induced ν -Gap	55
A.10.1 Bounded-influence training dynamics	56
A.10.2 A literature-supported route to persistent residual-motion drift	57
Bibliography	65

1

Introduction

Contents

1.1 Motivation	1
1.2 Project Aims and Contributions	2

1.1 Motivation

Distributed learning trains a model through cooperation between a server and a set of workers or clients. In a typical federated learning setting, the server keeps the global model, clients keep their data locally, and each communication round ends with the server aggregating the submitted updates [1], [2], [3]. At the same time, the aggregation step becomes a natural place for faulty or malicious clients to influence the global model.

Byzantine-robust aggregation addresses this risk by replacing simple averaging with rules that tolerate arbitrary or strategically malicious client messages [4], [5], [6]. This worst-case model poisoning view is important, but it is not the only practical threat. When client data are heterogeneous, some robust aggregators can become too conservative and may suppress useful non-IID updates [7].

The central problem is therefore a balance between robustness and tolerance of benign heterogeneity. Mean aggregation preserves heterogeneous updates, but it remains exposed to direct model-poisoning attacks. Hard filtering can block some strong attacks, but it can also remove

benign clients that naturally differ from the majority. History adds another part of the problem. Attacks that look small in one round can accumulate across time, while a client’s own past behaviour can provide a useful local reference [6], [8].

HRAC is designed for this middle ground. It limits the influence of large or unstable updates, but it does not treat every population-level deviation as malicious. Instead, it compares each client mainly with its own update trajectory before reducing the weight of unstable behaviour.

1.2 Project Aims and Contributions

This thesis designs and evaluates *HRAC* (History-Residual Adaptive Clipping), a centralised aggregation rule for this robustness and heterogeneity trade-off. HRAC combines an EMA-smoothed median norm cap, per-client history-residual clipping, and delayed soft weighting based on residual-change statistics. The history is client-specific. Past updates define a local reference for each client, rather than a single global centre shared by all clients. This allows HRAC to compare a client mainly with its own recent behaviour before applying bounded influence control.

The evaluation builds on the distributed learning code framework used by Peng et al. [7]. This project extends that framework with the HRAC aggregation rule, HRAC-specific diagnostic logging, and a broader attack suite used in the experiments reported here. Mean aggregation, robust baselines, and HRAC are compared on CIFAR-10 with IID and non-IID partitions and Byzantine poisoning settings. The evaluation uses fixed experimental settings and reports trends rather than full statistical benchmarks.

The remainder of the report is organised as follows. Chapter 2 reviews federated learning, Byzantine-robust aggregation, poisoning attacks, and history-aware defences. Chapter 3 presents the HRAC method and its implementation. Chapter 4 evaluates HRAC against baseline aggregators and reports component ablations. The conclusion discusses the main limitations and future work, and Appendix A gives the convergence analysis.

2

Background and Related Work

Contents

2.1	Federated Learning and Data Heterogeneity	3
2.2	Byzantine-Robust Aggregation	4
2.3	Poisoning Attacks and Attack-Dependent Robustness	6
2.4	History-Aware and Lightweight Defences	7
2.5	Research Gap	8

2.1 Federated Learning and Data Heterogeneity

Federated learning (FL) trains a shared model from decentralised data by iterating between local client optimisation and server-side aggregation. FedAvg and related model-averaging methods provide the standard starting point for this setting [1], [2]. Clients send training messages to the server rather than raw data, reducing central data collection and allowing training across distributed devices. Throughout this thesis, Δ_i^t denotes a gradient-like descent direction submitted by client i at round t . Thus the server moves in the negative direction of the aggregate. In the simplest server-side form, mean aggregation applies

$$g_t^{\text{mean}} = \frac{1}{N} \sum_{i=1}^N \Delta_i^t, \quad x_{t+1} = x_t - \eta g_t^{\text{mean}}. \quad (2.1)$$

This formula is useful but fragile. A single large or consistently biased Δ_i^t can shift the global direction. Production deployments must handle partial participation, client availability, stragglers,

communication limits, and privacy-preserving aggregation protocols [9], [10]. Benchmarking work such as LEAF also emphasises that realistic federated data are naturally heterogeneous across users and tasks [11].

The main complication for robust aggregation is that realistic FL is rarely IID. Client datasets can be label-skewed and unbalanced, and they are often shaped by local usage patterns [3]. As a result, benign client messages may legitimately differ in direction and magnitude. Optimisation work such as FedProx [12], SCAFFOLD [13], and adaptive server methods [14] address benign instability under heterogeneity, but these methods do not by themselves protect the server from malicious updates. Empirical studies of federated visual classification further show that the degree of non-IIDness can materially change FedAvg behaviour [15]. For this paper, heterogeneity is not only an optimisation difficulty, because benign diversity can be mistaken for adversarial behaviour. Recent Byzantine-robust FL work therefore treats heterogeneity as part of the security problem rather than as a secondary nuisance. Resampling and mixing strategies have been proposed to reduce the gap between benign heterogeneous clients before applying robust aggregation [16], [17]. HRAC follows the same broad motivation, but it avoids constructing mixed client batches and instead keeps a lightweight history for each individual client.

2.2 Byzantine-Robust Aggregation

Byzantine robustness studies aggregation when an unknown subset of workers may send arbitrary or coordinated messages [4], [18]. In distributed and federated learning, the server usually sees only the submitted updates, not the clients' data or intentions. The aggregation rule is therefore the main defence against faulty or malicious clients.

Existing defences can be grouped by their aggregation principle. Selection-based rules such as Krum choose updates that appear close to other updates [5]. In one common description, Krum scores client i by the sum of squared distances to its $N - b - 2$ nearest neighbours and selects the lowest-scoring update:

$$s_i = \sum_{j \in \mathcal{N}_i} \|\Delta_i^t - \Delta_j^t\|_2^2, \quad g_t^{\text{Krum}} = \Delta_{i^*}^t, \quad i^* = \arg \min_i s_i. \quad (2.2)$$

Filtering-based methods use a related distance-based idea but remove suspected outliers before averaging the remaining updates. FABA, for example, repeatedly removes the update farthest from the current aggregate estimate and then averages the surviving updates [19]. Coordinate-wise

methods such as median and trimmed mean apply robust statistics independently by coordinate [20]. For coordinate k , these rules can be written as

$$[g_t^{\text{med}}]_k = \text{median}_i [\Delta_i^t]_k, \quad [g_t^{\text{trim}}]_k = \frac{1}{N - 2b} \sum_{r=b+1}^{N-b} z_{(r),k}, \quad (2.3)$$

where $z_{(r),k}$ are the sorted values of $\{[\Delta_i^t]_k\}_{i=1}^N$. High-dimensional robust estimation methods, including geometric-median-based RFA [21], estimate a central update with reduced sensitivity to outliers. Centered clipping instead treats clipping as influence control rather than attacker removal [6]. The Byzantine message may still enter the average, but only through a bounded residual around a reference point. Its analysis is most natural when honest updates concentrate around a common reference, which is closer to an IID or weakly heterogeneous picture than to strongly label-skewed FL. A typical centred-clipping step clips each message around a reference c_t :

$$\tilde{\Delta}_i^t = c_t + \min\left(1, \frac{\tau}{\|\Delta_i^t - c_t\|_2}\right) (\Delta_i^t - c_t), \quad g_t = \frac{1}{N} \sum_i \tilde{\Delta}_i^t. \quad (2.4)$$

Other lines of work directly address robustness under heterogeneous data, such as RSA, which incorporates a regularisation-based robust aggregation objective for non-IID distributed datasets [22]. Outside strictly Byzantine settings, adaptive clipping has also been used in private FL to tune clipping thresholds from update-norm statistics rather than from expensive manual search [23]. This line of work supports the use of norm-based, data-dependent caps, although its goal is privacy accounting rather than adversarial robustness.

Together, these methods show that robustness often comes from reducing the effect of non-central updates. The limitation is that benign clients may also be non-central under heterogeneous data. A single centre can therefore be a poor reference for legitimate label-driven differences. The experiments in Chapter 4 return to this issue for centred clipping. Selection-based rules also have known vulnerabilities under coordinated attacks [24]. Variance-based attacks add another difficulty because malicious updates can look normal under benign update statistics while still damaging training [8]. Recent robust-estimation work also notes a sample-efficiency trade-off, where robust rules may need more participating clients than averaging to recover the same utility in clean settings [25]. These observations motivate robust methods that limit harmful updates without automatically penalising every client that differs from the population centre.

2.3 Poisoning Attacks and Attack-Dependent Robustness

Poisoning attacks can target the submitted descent direction directly or corrupt local data so that protocol-following training produces harmful messages [3]. Model-poisoning attacks are often stronger because the attacker can craft the submitted message itself. Some attacks are designed specifically to degrade learning while evading robust aggregation rules [26]. Production-oriented evaluations further show that the damage caused by poisoning depends on deployment assumptions, threat-model realism, and how the attack interacts with the aggregation rule [27]. This makes it important to report not only final accuracy, but also the conditions under which a defence is expected to work.

Stealthy attacks are especially relevant to this project because they challenge simple outlier assumptions. ALIE exploits benign variance so that malicious updates remain within a statistically plausible range [8]. If μ and σ are coordinate-wise statistics of the benign updates, ALIE can submit

$$\Delta_{\text{ALIE}} = \mu - z\sigma, \quad (2.5)$$

for a scalar z chosen so that the malicious update is shifted but not obviously extreme in each coordinate. IPM and related sign or direction manipulations can push the aggregate away from the benign direction without requiring obviously large norms; a simple IPM form is

$$\Delta_{\text{IPM}} = -\epsilon \bar{\Delta}_{\text{benign}}, \quad (2.6)$$

where $\bar{\Delta}_{\text{benign}}$ is the benign mean direction and $\epsilon > 0$ controls attack strength. Model Shuffle Attack (MSA) further manipulates parameter structure by shuffling locations and scaling updates, degrading differential-privacy-based FL while avoiding naive anomaly checks [28]. These attacks suggest that robustness cannot depend only on detecting extreme update magnitudes.

Data poisoning represents a protocol-compliant and often realistic threat model. In label flipping, a malicious client changes part of its local labels and then follows the normal training protocol. The submitted message can therefore look plausible because it is produced by standard local optimisation rather than by direct message fabrication. Label-poisoning attacks and defences have been studied in FL settings [29], [30]. Formally, the worker still sends a protocol-compliant

message, but it is computed from a poisoned local objective, for example

$$\tilde{f}_i(x) = \frac{1}{|\mathcal{D}_i|} \sum_{(u,y) \in \mathcal{D}_i} \ell(x; u, \pi(y)), \quad \Delta_i^t \approx \nabla \tilde{f}_i(x_t), \quad (2.7)$$

where π denotes a label-flipping map. This explains why label poisoning can be difficult for geometry-only filters: the message is harmful but is still generated by ordinary local training.

Recent work makes this point sharply for label poisoning: robust aggregators designed for arbitrary model poisoning can become overly conservative, and may underperform when the data are sufficiently heterogeneous [7]. Gradient-space methods such as LFighter attempt to detect poisoned clients by clustering worker gradients [31], but clustering can become unreliable when benign data are strongly non-IID. HRAC takes two lessons from these results. First, hard filtering should be avoided when benign heterogeneous updates may be useful. Second, robustness has to be reported with the attack model and the degree of client heterogeneity, rather than under a single Byzantine abstraction.

2.4 History-Aware and Lightweight Defences

History is useful in distributed learning because it can stabilise noisy optimisation. Server-side momentum and adaptive optimisation use temporal state to improve convergence under stochasticity and partial participation [14]. A representative server momentum recursion is

$$m_t = \beta m_{t-1} + (1 - \beta) g_t, \quad x_{t+1} = x_t - \eta m_t. \quad (2.8)$$

Robust optimisation can also exploit history: learning-from-history frameworks construct robust directions from past information, and centered clipping with momentum demonstrates that historical state can improve Byzantine resilience [6]. Variance-reduced and momentum-based robust methods similarly argue that reducing stochastic variation can make malicious messages easier to separate from noisy benign updates [32], [33]. Historical-gradient studies also examine whether storing past client gradients helps detect attacks, but they highlight the storage and privacy costs of keeping high-dimensional historical update records [34]. This motivates compact temporal summaries rather than full-gradient histories.

A malicious client may apply small but persistent perturbations that are inconspicuous in a single round but accumulate over time. HisMSA explicitly models historical knowledge to craft

poisoning updates that bypass multiple countermeasures [35]. This creates a design tension: a defence should use history, but it should do so in a way that detects persistent residual change rather than blindly trusting historical momentum.

Deployability adds a further constraint to history-aware robustness. Practical FL systems may use privacy-enhancing mechanisms such as secure aggregation. In this setting, the server recovers only an aggregate sum rather than each individual update [36]. This constrains robust defences, since many anomaly detectors require detailed per-client visibility. Other approaches, such as FLTrust, use a server-side reference update to reweight client messages by alignment [37]. Such methods improve robustness but introduce additional assumptions about the server’s trusted data or reference direction.

Scalability also matters. Pairwise distances, clustering, and repeated high-dimensional comparisons can become expensive as the number of clients and model dimension increase. Lightweight per-client statistics, such as update norms, residual scales, and residual-change summaries, offer a practical alternative. They can be maintained with exponential moving averages and used to detect abrupt or persistent behavioural changes without repeatedly solving a large robust-estimation problem.

2.5 Research Gap

The related work motivates a narrower problem than simply “be robust to all Byzantine clients”. Worst-case robust aggregators can be effective against arbitrary model poisoning, but under heterogeneous data they may suppress benign clients that naturally differ from the majority. Mean aggregation has the opposite weakness. It is less conservative, but it remains exposed to stronger model-poisoning attacks.

HRAC is positioned between these two behaviours. It keeps an explicit norm cap for strong malicious updates, but it does not rely only on distance from a global population centre. This is where it departs from centre-based clipping. The global cap controls the scale of each submitted message, while the residual test compares each client mainly with its own historical behaviour. Clients whose residual dynamics become unstable are penalised softly rather than removed outright.

3

Method and Implementation

Contents

3.1 Design Rationale	9
3.2 Problem Setup and Threat Model	10
3.3 Aggregation Pipeline	11
3.4 Core Components	11
3.4.1 Global Median Norm Cap	11
3.4.2 Per-Client History-Residual Clipping	13
3.4.3 Separated Aggregation and Statistics Paths	13
3.4.4 EMA History Updates	14
3.4.5 Soft Weighting from the ν Statistic	14
3.5 Software Integration and Logging	15
3.6 Hyperparameters	16
3.7 Convergence Statement	16

3.1 Design Rationale

The design follows from three limitations in the related work. Mean aggregation tolerates heterogeneous benign clients, but a large or coordinated model-poisoning update enters the average with linear influence. Population-centred robust rules, including centred clipping, reduce this risk by comparing clients with a common centre, but that centre can be misleading under non-IID data. Temporal methods show that history can stabilise learning, but the historical reference should reflect each client’s own behaviour rather than one global pattern shared by all clients.

These observations determine the HRAC components. Like centred clipping, HRAC uses clipping to bound influence rather than to make Byzantine updates vanish. The global median norm

cap limits large-scale malicious updates before they enter the aggregate. The history-residual clipping step then allows each client to affect the round only through a bounded residual relative to its own recent behaviour, so a client is not penalised only because it is far from the population centre. The ν_i statistic tracks residual movement over time, and the weight penalty suppresses persistently elevated movement gradually before the weighted aggregation. HRAC is therefore not a hard detector. It is an influence-limiting rule designed for settings where benign heterogeneity and malicious update deviations can coexist.

3.2 Problem Setup and Threat Model

Consider a federated learning system with N clients and a global server. At communication round t , each participating client computes a local training message $\Delta_i^t \in \mathbb{R}^d$ and sends it to the server. As in Chapter 2, Δ_i^t is treated as a gradient-like descent direction, so the global model is updated by subtracting the final aggregate. The server applies an aggregation rule

$$\text{Agg}(\Delta_1^t, \dots, \Delta_N^t)$$

to form a single update g_t , which is then used to update the global model.

The client population is divided into an unknown benign set $\mathcal{G}$ and a Byzantine set $\mathcal{B}$, with $|\mathcal{B}| = b$ and $|\mathcal{G}| = N - b$. Clients in $\mathcal{G}$ follow the training protocol, but their updates may be heterogeneous because their local data are non-IID. Clients in $\mathcal{B}$ may send arbitrary or strategically crafted messages. In the experiments, this includes both direct model manipulation attacks, such as bit flipping, ALIE, IPM, MSA, and HisMSA, and protocol-compliant data poisoning through label flipping.

The objective is not to perfectly classify clients as benign or malicious. Instead, the goal is to limit the influence of harmful updates while preserving useful contributions from benign but heterogeneous clients. This distinction is important under non-IID data: a benign client may be far from the population centre for legitimate statistical reasons, so hard outlier removal can damage utility.

HRAC is designed for this setting with two assumptions. First, extreme round-wise update magnitudes should be bounded before aggregation. Second, client behaviour should be compared primarily with the client’s own historical trajectory rather than only with a global population

centre.

3.3 Aggregation Pipeline

In communication round t , the server receives flattened client messages $\{\Delta_i^t\}_{i=1}^N$, with $\Delta_i^t \in \mathbb{R}^d$. HRAC first limits extreme magnitudes using a median-based cap, then processes each client relative to its own residual history. The clipped residuals are used to score cross-round movement through ν_i . After a warm-up period, the server turns these scores into soft weights, forms the weighted aggregate, and then updates the per-client histories.

Algorithm 1 summarises the complete server-side HRAC procedure. The rest of this section gives the definitions and design rationale behind each stage.

3.4 Core Components

3.4.1 Global Median Norm Cap

The first defence layer prevents a single large update from dominating the aggregate. HRAC smooths the round-wise median update norm with an EMA before forming the clipping threshold. Let

$$m_t = \text{med}_{1 \leq j \leq N} \|\Delta_j^t\|_2, \quad \bar{m}_t = \rho_g \bar{m}_{t-1} + (1 - \rho_g) m_t.$$

The clipping threshold used in round t is formed from the stored scale at the start of the round,

$$B_t = c_g \max\{\bar{m}_{t-1}, s_{\min}\},$$

where $s_{\min} = 10^{-3}$ in the implementation. The first round warm-starts $\bar{m}$ from the current median norm. Thus, the current round contributes to the next round's cap rather than immediately changing its own threshold. This makes the cap predictable with respect to the current batch of submitted messages and reduces sensitivity to one-round norm spikes or drops. Each message is clipped by

$$\hat{\Delta}_i^t = \min\left(1, \frac{B_t}{\|\Delta_i^t\|_2 + \epsilon}\right) \Delta_i^t. \quad (3.1)$$

The implementation uses $c_g = 3.0$ and $\rho_g = 0.95$ by default. The cap remains median-based, so it does not require knowing which clients are malicious and is less sensitive to extreme Byzantine

Algorithm 1 History-Residual Adaptive Clipping (HRAC)

Require: messages $\{\Delta_i^t\}_{i=1}^N$, client ids, states $(b_i, \mu_i, \nu_i, p_i, \bar{r}_i^{\text{prev}}, \bar{m})$, parameters $c_g, c, \rho_g, \rho_b, \rho_\mu, \rho_\nu, \rho_p, T_\nu, \alpha, w_{\max}$

Ensure: aggregated update g_t

- 1: $m_t \leftarrow \text{med}_j \|\Delta_j^t\|_2$
- 2: if $\bar{m}$ is uninitialised, set $\bar{m} \leftarrow m_t$; $B_t \leftarrow c_g \max(\bar{m}, s_{\min})$
- 3: **for** each client i **do**
- 4: $\hat{\Delta}_i^t \leftarrow \min(1, B_t/(\|\Delta_i^t\|_2 + \epsilon))\Delta_i^t$
- 5: **if** i is a new client **then**
- 6: $b_i \leftarrow \hat{\Delta}_i^t$; $\mu_i, \nu_i \leftarrow \min(\max(m_t, \mu_{\min}), 2B_t)$
- 7: $p_i \leftarrow 0$; $\bar{r}_i^{\text{prev}} \leftarrow 0$
- 8: $\tilde{\Delta}_i^t \leftarrow \hat{\Delta}_i^t$
- 9: **else**
- 10: $r_i^t \leftarrow \hat{\Delta}_i^t - b_i$; $\tau_i^t \leftarrow c\mu_i + \epsilon$
- 11: $\bar{r}_i^t \leftarrow \min(1, \tau_i^t/(\|r_i^t\|_2 + \epsilon))r_i^t$
- 12: $\tilde{\Delta}_i^t \leftarrow b_i + \bar{r}_i^t$
- 13: **end if**
- 14: **end for**
- 15: **if** $t \leq T_\nu$ **then**
- 16: $w_i^t \leftarrow 1/N$ for all clients
- 17: **else**
- 18: $\bar{\nu}^t \leftarrow N^{-1} \sum_{j=1}^N \nu_j$
- 19: **for** each client i **do**
- 20: $e_i^t \leftarrow \mathbb{I}[\nu_i > 0.9] \max(\nu_i - \bar{\nu}^t, 0)$
- 21: $p_i \leftarrow \rho_p p_i + (1 - \rho_p)e_i^t$
- 22: $\hat{w}_i^t \leftarrow 1/(1 + \alpha p_i)$
- 23: **end for**
- 24: normalise $\{\hat{w}_i^t\}$ and cap weights by $w_{\max}$
- 25: **end if**
- 26: $g_t \leftarrow \sum_{i=1}^N w_i^t \tilde{\Delta}_i^t$
- 27: **for** each non-new client i **do**
- 28: $\tilde{\Delta}_{i,\text{safe}}^t \leftarrow \tilde{\Delta}_i^t$
- 29: $b_i \leftarrow \rho_b b_i + (1 - \rho_b)\tilde{\Delta}_{i,\text{safe}}^t$
- 30: $\mu_i \leftarrow \rho_\mu \mu_i + (1 - \rho_\mu)\|\bar{r}_i^t\|_2$
- 31: $d_i^t \leftarrow \|\bar{r}_i^t - \bar{r}_i^{\text{prev}}\|_2$
- 32: $\nu_i \leftarrow \rho_\nu \nu_i + (1 - \rho_\nu)d_i^t$; $\bar{r}_i^{\text{prev}} \leftarrow \bar{r}_i^t$
- 33: **end for**
- 34: $\bar{m} \leftarrow \rho_g \bar{m} + (1 - \rho_g)m_t$
- 35: **return** g_t

norms than a mean-based scale estimate; the additional EMA smoothing makes the cap less volatile across communication rounds.

3.4.2 Per-Client History-Residual Clipping

For each client i , HRAC stores a historical centre b_i^t . This centre is an EMA of safe past messages and represents the client’s own typical update path, not a population-wide prototype. The residual is

$$r_i^t = \hat{\Delta}_i^t - b_i^t. \quad (3.2)$$

The residual threshold is adaptive:

$$\tau_i^t = c\mu_i^t + \epsilon, \quad (3.3)$$

where μ_i^t is an EMA scale statistic and $c = 2.5$ by default. The clipped residual is

$$\bar{r}_i^t = \min\left(1, \frac{\tau_i^t}{\|r_i^t\|_2 + \epsilon}\right) r_i^t. \quad (3.4)$$

The processed message is then reconstructed as

$$\tilde{\Delta}_i^t = b_i^t + \bar{r}_i^t. \quad (3.5)$$

This reconstructed message is used directly for aggregation after residual clipping. In the reported runs, the same reconstructed message is also used to update the client history centre, so

$$\tilde{\Delta}_{i,\text{safe}}^t = \tilde{\Delta}_i^t.$$

3.4.3 Separated Aggregation and Statistics Paths

The implementation distinguishes the aggregation path from the statistics path. Aggregation uses the globally clipped update $\hat{\Delta}_i^t$ and its processed version $\tilde{\Delta}_i^t$. The statistics path can lift extremely small-norm updates to the median norm when computing μ_i and ν_i . This prevents a near-zero update from artificially collapsing a client’s history statistics. The aggregation value itself is not lifted in this case, so the actual descent direction remains faithful to the received message after global clipping.

3.4.4 EMA History Updates

After aggregation, HRAC updates the client histories. The aggregation-path centre is

$$b_i^{t+1} = \rho_b b_i^t + (1 - \rho_b) \tilde{\Delta}_{i,\text{safe}}^t, \quad (3.6)$$

where $\tilde{\Delta}_{i,\text{safe}}^t$ is the message used in the history update. For the reported runs, this is the same reconstructed message used in the aggregate. The default value is $\rho_b = 0.98$.

The residual scale statistic is updated as

$$\mu_i^{t+1} = \rho_\mu \mu_i^t + (1 - \rho_\mu) \|\bar{r}_i^t\|_2, \quad (3.7)$$

with a small floor $\mu_{\min}$ for numerical stability. The default is $\rho_\mu = 0.95$.

The change statistic ν_i is updated from the movement of the clipped residual:

$$d_i^t = \|\bar{r}_i^t - \bar{r}_i^{t-1}\|_2, \quad \nu_i^{t+1} = \rho_\nu \nu_i^t + (1 - \rho_\nu) d_i^t, \quad (3.8)$$

again with a small floor $\nu_{\min}$. The class default is $\rho_\nu = 0.95$. In the reported experiment driver, this value is set to $\rho_\nu = 0.87$ so that the residual-change statistic reacts slightly faster to persistent movement.

3.4.5 Soft Weighting from the ν Statistic

The weighting rule is deliberately delayed. HRAC does not immediately down-weight clients from the first round. For iterations $t \leq T_\nu$, it uses uniform weights. In the implementation, $T_\nu = 50$ by default. This warm-up allows the per-client statistics to become meaningful before they affect aggregation.

Once the penalty is active, the server forms a population baseline

$$\bar{\nu}^t = \frac{1}{N} \sum_{j=1}^N \nu_j^t. \quad (3.9)$$

Only clients with sufficiently high ν_i^t are penalised. In code, the activation threshold is 0.9, and

the excess score is

$$e_i^t = \begin{cases} \max(\nu_i^t - \bar{\nu}^t, 0), & \nu_i^t > 0.9, \\ 0, & \nu_i^t \leq 0.9. \end{cases} \quad (3.10)$$

The excess score is itself smoothed before being used for aggregation:

$$p_i^t = \rho_p p_i^{t-1} + (1 - \rho_p) e_i^t. \quad (3.11)$$

The implementation uses $\rho_p = 0.90$. This extra EMA prevents a single large residual-change measurement from causing an abrupt weight change. The raw weight is then

$$\hat{w}_i^t = \frac{1}{1 + \alpha p_i^t}, \quad (3.12)$$

where $\alpha = 5.0$ by default. Raw weights are normalised to sum to one. A maximum per-client weight cap, $w_{\max} = 0.30$, is then applied iteratively with renormalisation to avoid a small number of clients dominating the round. More explicitly, let $q_i = \hat{w}_i^t / \sum_k \hat{w}_k^t$. If all $q_i \leq w_{\max}$, then $w_i^t = q_i$. Otherwise, all clients with $q_i > w_{\max}$ are fixed at $w_{\max}$; the remaining probability mass is renormalised over the uncapped clients in proportion to their current q_i values. This fix-and-renormalise step is repeated until no weight exceeds $w_{\max}$. The resulting weights are the final aggregation weights used in g_t . In the convergence analysis, quantities such as the Byzantine weight mass β_t are defined after this capping step.

The final aggregate is

$$g_t = \sum_{i=1}^N w_i^t \tilde{\Delta}_i^t. \quad (3.13)$$

3.5 Software Integration and Logging

HRAC is implemented as the `C_HRAC` aggregation class in the same centralised aggregation interface used by the baseline rules. The training loop therefore remains the same across mean aggregation, FABA, centred clipping, LFighter, and HRAC. The same experiment driver also constructs the attack object, including label flipping, bit flipping, ALIE, IPM, MSA, and HisMSA, so the method is evaluated under matched training settings rather than through attack-specific code.

The ablation experiments use controlled variants of the same HRAC pipeline by removing the global median cap, per-client residual clipping, ν -based weighting, or both history-based stages.

HRAC also records diagnostic quantities at a fixed logging interval, including residual thresholds, μ and ν statistics, pre- and post-clipping norms, normalised weights, and aggregate norms. Accuracy, loss, metadata, and diagnostic logs are stored through the common cache interface and reused by the plotting scripts in Chapter 4.

3.6 Hyperparameters

The main HRAC hyperparameters are:

- $\rho_b = 0.98$: EMA decay for the client history centre.
- $\rho_\mu = 0.95$: EMA decay for the residual scale statistic.
- $\rho_\nu = 0.87$ in the experiment driver: EMA decay for the residual-change statistic. The class default is 0.95, but the reported runs use a slightly faster ν update.
- $c = 2.5$: multiplier for the per-client residual clipping threshold.
- $c_g = 3.0$: multiplier for the EMA-smoothed global median norm cap.
- $\rho_g = 0.95$: EMA decay for the global median-norm scale.
- $T_\nu = 50$: first iteration after which ν -based weighting is used.
- $\alpha = 5.0$: strength of the ν excess penalty.
- $\rho_p = 0.90$: EMA decay for the smoothed ν -penalty score.
- $w_{\max} = 0.30$: maximum normalised weight allowed for one client.

The algorithm is intentionally not a hard detector. Its most sensitive parameters are c , which controls how much residual deviation is allowed before clipping, and α , which controls how sharply elevated ν values reduce weights.

3.7 Convergence Statement

The empirical evaluation in Chapter 4 is the primary evidence for HRAC’s practical robustness. The method can also be analysed through a modular convergence argument. The analysis treats

HRAC as an aggregation rule that produces g_t , compares this aggregate with the honest conditional centre, and inserts the resulting aggregation error into the standard descent argument for smooth non-convex optimisation. The lagged EMA-smoothed median cap makes the clipping radius predictable before the current round’s stochastic updates are observed. Under additional admissibility conditions on the honest conditional centre and residual scale, this allows the clipped-moment lemma of Sadiev et al. [38, Lemma 5.1] to be applied.

Under bounded honest updates, an honest-majority condition for the median-based scale, bounded honest clipping distortion, bounded honest-weight distortion, and bounded residual Byzantine weight mass, this conditional analysis shows that HRAC converges to a first-order stationary neighbourhood in the averaged-gradient sense. In particular, for a sufficiently small constant step size η , the averaged squared gradient norm satisfies a bound of the form

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla f(x_t)\|^2 \leq \frac{2(f(x_0) - f^*)}{\eta T} + R_{\text{HRAC}}, \quad (3.14)$$

where R_{HRAC} collects the honest stochastic variance, non-IID/bias term, global clipping distortion, residual clipping distortion, and the remaining Byzantine aggregation error. When the logged dynamics exhibit a late-stage group-level ν -gap, the weighting rule can reduce the residual Byzantine mass appearing in this error term.

This result should be read as a conditional bounded-influence stability statement rather than as a claim of exact convergence to a zero-gradient point under arbitrary attacks. The proof does not establish universal Byzantine identification. Persistent heterogeneity and nonzero Byzantine mass create a nonzero error floor. The full proof, including the aggregation-error decomposition and the explicit neighbourhood constant, is given in Appendix A.

4

Experiments and Results

Contents

4.1 Experimental Setup	19
4.2 Experimental Results	20
4.3 Additional Analyses	23
4.3.1 Final and Best Accuracy Summary	23
4.3.2 Component Ablation Study	27
4.3.3 Diagnostic Analysis	28
4.4 Discussion and Limitations	30

4.1 Experimental Setup

The experiments are intended to evaluate both benign utility and Byzantine robustness under the same implementation pipeline. The training code is based on the distributed learning framework used by Peng et al. [7]. This project adds HRAC as a new server-side aggregation rule, implements the attack settings used in this study, and records the HRAC diagnostics used later in this chapter. Table 4.1 summarises the common settings used for the CIFAR-10 results. The baseline condition uses $b = 0$ to verify that HRAC does not substantially harm normal training. The main attack condition uses $b = 3$ out of $N = 10$ clients, corresponding to 30% malicious participation. A supplementary set of figures and summary values is also reported for $b = 2$, corresponding to 20% malicious participation.

Table 4.2 makes the experimental scope explicit. The reported curves should be read as fixed-seed comparisons, not as confidence-interval benchmarks. The attack suite includes both structured

Table 4.1: Experimental setup for the CIFAR-10 evaluation.

Item	Setting
Dataset	CIFAR-10
Model	Small CNN with two convolutional layers, adaptive pooling, and two fully connected layers
Client population	$N = 10$ clients
Byzantine level	$b = 3$ main attack setting; $b = 2$ supplementary attack setting; $b = 0$ benign baseline
Data partitions	IID random split and non-IID label-skew split, with one CIFAR-10 label class per client in the non-IID case
Training horizon	20,000 communication iterations; results are logged every 100 iterations, giving 200 plotted/evaluated points
Attack settings	Label flipping; MSA/HisMSA with shuffle probability 1.0 and scaling range (0.1, 10.0); bit flipping; ALIE; IPM with $\epsilon = 0.1$
Aggregation baselines	Mean, FABAs, centred clipping (CC), HRAC, and LFighter in the label-flipping curves where cached runs are available
Main metric	Test accuracy over communication iterations
Diagnostics	Residual thresholds, μ/ν statistics, normalised weights, and bottom-weight summaries in controlled runs

Table 4.2: Coverage and scope of the reported experiments.

Dimension	Covered in this report	Not covered systematically
Dataset	CIFAR-10	Additional datasets and model families
Client population	$N = 10$ clients	Larger or varying client populations
Byzantine fraction	$b = 3$ main setting and $b = 2$ supplementary setting; $b = 0$ benign baseline	Broader sweeps over attacker fractions
Data heterogeneity	IID and non-IID partitions	Multiple non-IID severity levels
Randomness	Fixed cached runs used consistently across methods	Multi-seed confidence intervals
Hyperparameters	One fixed HRAC configuration across main runs	Systematic hyperparameter sweeps

model-poisoning attacks and simpler settings such as label flipping and bit flipping, since recent work shows that aggregator rankings can change with the attack model and data heterogeneity [7]. Sensitivity to seeds, larger client populations, and different Byzantine fractions remains future work.

All methods are compared using the same cached training logs and plotting pipeline. The figures report accuracy trajectories as well as final values because some attacks cause early instability, while history-aware attacks can produce slower cross-round drift. The summary tables report final and best test accuracy. The best value is used as a supplementary stability indicator, while the discussion is based mainly on final accuracy and the trajectory plots.

4.2 Experimental Results

Figures 4.1–4.3 summarise the main CIFAR-10 experiments used in this report. The plots compare HRAC against mean aggregation and robust baselines under IID and non-IID client partitions. The training setup uses 10 clients, and the main attack figures use $b = 3$, corresponding to 30% malicious participants. The horizontal axis reports communication iterations, and the vertical axis reports test accuracy. The non-IID partition follows the setting used by the implementation logs, where clients observe different subsets of CIFAR-10 labels. LFighter is included in the label-flipping panels, where cached runs are available for both $b = 2$ and $b = 3$.

Figure 4.1 first checks benign utility and a protocol-compliant poisoning setting. Under benign training ($b = 0$), HRAC tracks the strongest baselines closely, indicating that the proposed aggregation rule does not introduce a large utility penalty when all clients are honest. Under label flipping, HRAC improves the early IID trajectory and remains competitive, but FABA, LFighter, and mean aggregation can be close or better in some regimes, especially under non-IID data. LFighter is close to the strongest robust baseline under IID label flipping, but its non-IID label-flipping result is weaker than mean aggregation in these cached runs. This case differs from direct model manipulation because poisoned clients can still produce updates from the normal local training procedure.

Figure 4.2 shows attacks that manipulate update structure or exploit temporal state. In the MSA panels, mean aggregation degrades severely, especially under non-IID data, whereas HRAC maintains a much higher trajectory. In the HisMSA panels, HRAC remains close to the best curves and is particularly stable in the non-IID setting, which is consistent with the design motivation for using client-specific history. These results show that HRAC is most effective when the attack creates structured or persistent deviation rather than only noisy benign variation.

Figure 4.3 reports bit flipping, ALIE, and IPM. The IID results are relatively close across several methods, but the non-IID panels separate the defences more clearly. HRAC is consistently among the strongest and most stable methods for non-IID ALIE and IPM. Bit flipping is more mixed in the final-accuracy summaries: HRAC clearly suppresses the attackers in the diagnostic plots, but FABA is higher by about five percentage points in the non-IID $b = 3$ final/best accuracy table. The centred-clipping baseline also illustrates the limitation discussed in Chapter 2: even when used in the same baseline aggregation setting, a single centre is not a reliable reference under the non-IID partition. Overall, the results support the central design claim of HRAC: comparing

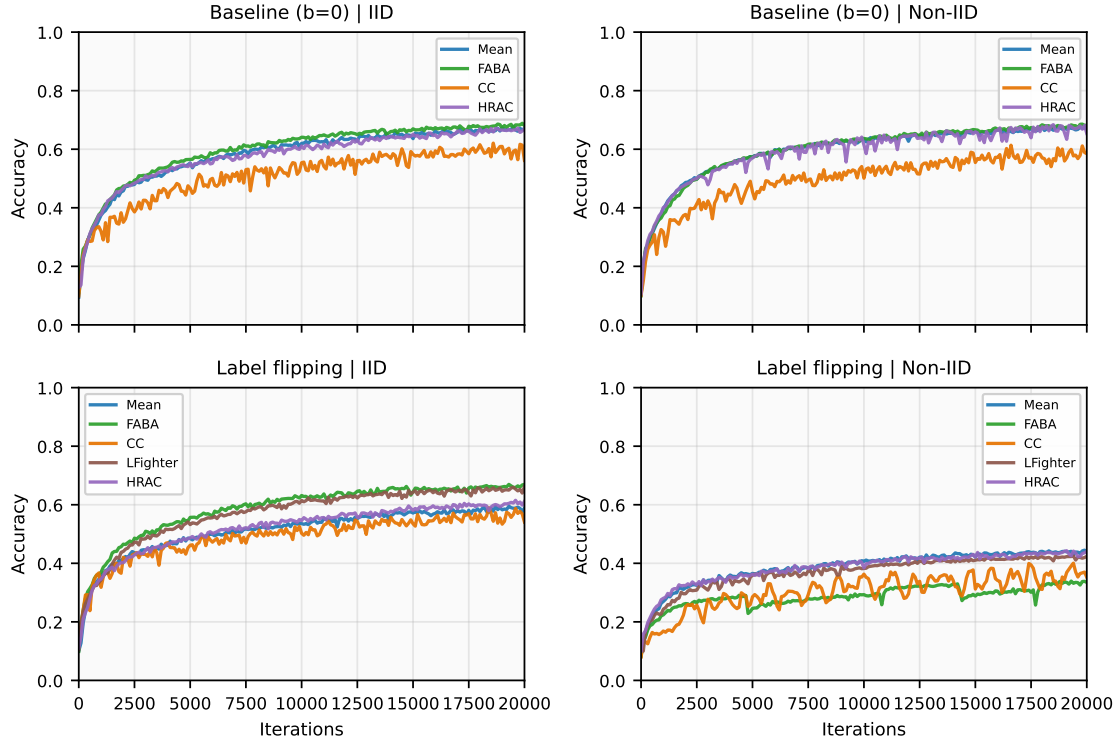


Figure 4.1: CIFAR-10 accuracy under benign training and label-flipping poisoning. The top row uses $b = 0$; the label-flipping row uses $b = 3$.

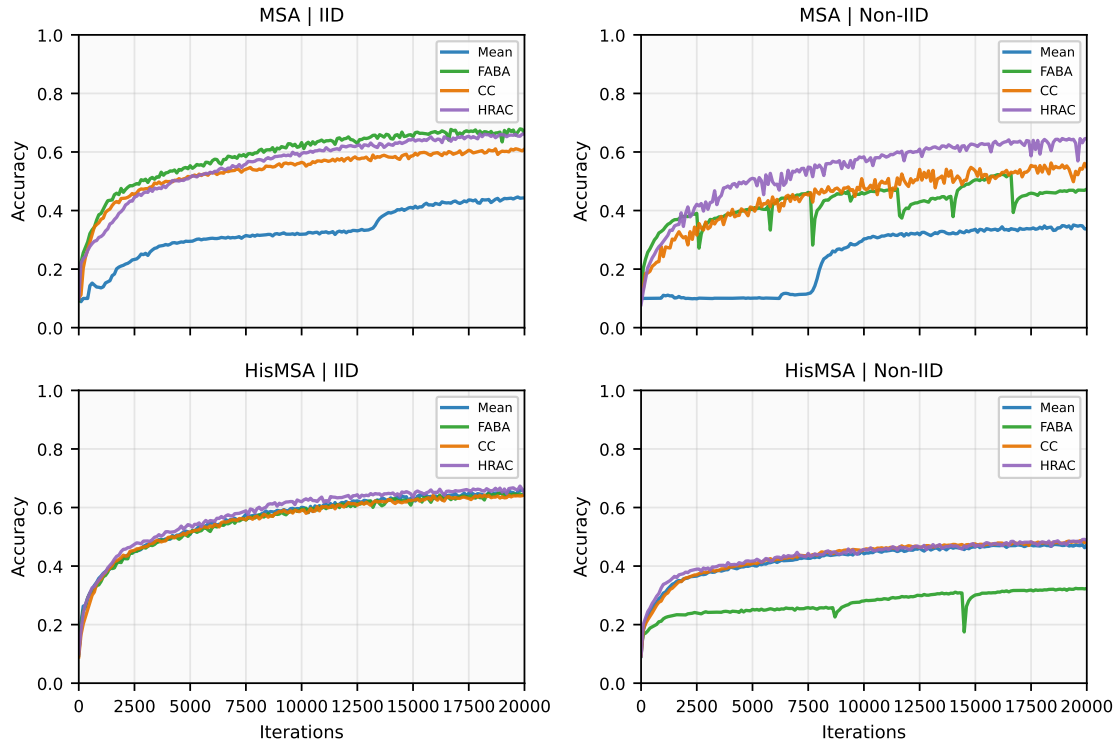


Figure 4.2: CIFAR-10 accuracy under model-shuffling (MSA) and history-aware model-shuffling (HisMSA) attacks with $b = 3$.

clients against their own histories can improve robustness to structured and history-aware attacks while preserving benign training behaviour. At the same time, the mixed label-flipping and bit-flipping panels show that HRAC should be treated as a robust aggregation strategy rather than a universally dominant detector.

Figures 4.4–4.6 repeat the same comparison with $b = 2$. The lower attacker fraction reduces the severity of several attacks, especially label flipping and ALIE, but the broad pattern remains attack-dependent. Mean aggregation can recover in some protocol-compliant data-poisoning settings. LFighter again remains competitive for IID label flipping but does not resolve the non-IID label-flipping case. HRAC remains strongest in the structured MSA setting and competitive in the remaining attacks.

4.3 Additional Analyses

Beyond the main comparison curves, the implementation was also used to isolate which parts of HRAC are responsible for the observed behaviour. The component study below is reported for non-IID CIFAR-10, where benign heterogeneity makes hard filtering risky and where the main robustness claims are most relevant. The remaining analysis focuses on what can be concluded from the available logs, while broader sensitivity studies are treated as limitations and future work.

4.3.1 Final and Best Accuracy Summary

Tables 4.3 and 4.4 complement the accuracy curves by reporting final-round and best observed test accuracy. Each entry is written as “final / best”. The best robust baseline column is selected among FABA and CC, excluding both mean aggregation and HRAC, so that mean aggregation remains visible as a separate reference. LFighter is shown in the label-flipping curves but is not included in these summary tables because cached LFighter runs are available only for label flipping, not for the full attack suite.

4.3.2 Component Ablation Study

The ablation study removes individual HRAC components while keeping the same training setting. The tested variants remove the global median cap, remove per-client residual clipping, remove ν -based weighting, or keep only the global cap before averaging. The goal is to check whether

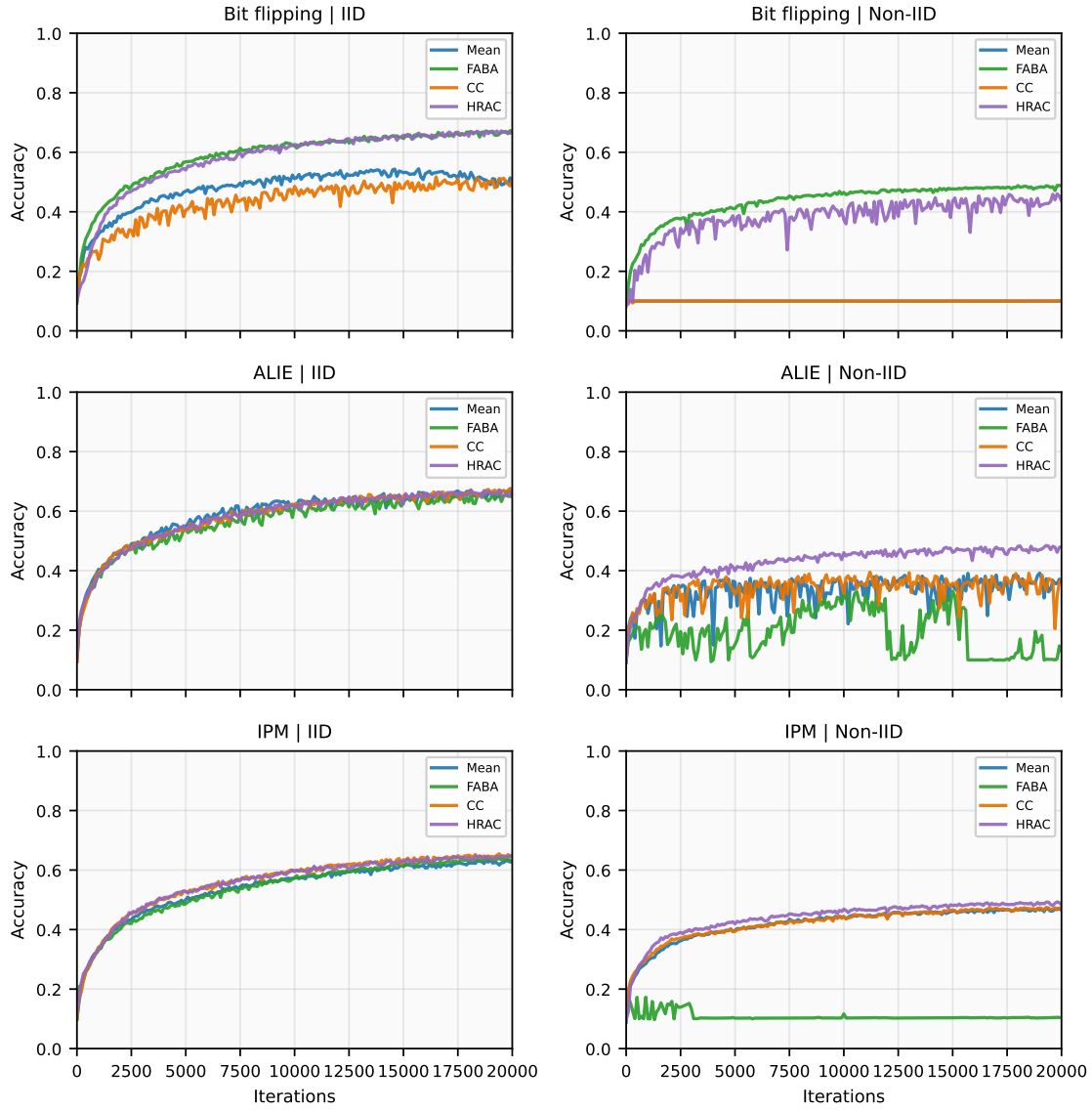


Figure 4.3: CIFAR-10 accuracy under bit-flipping, ALIE, and IPM attacks with $b = 3$.

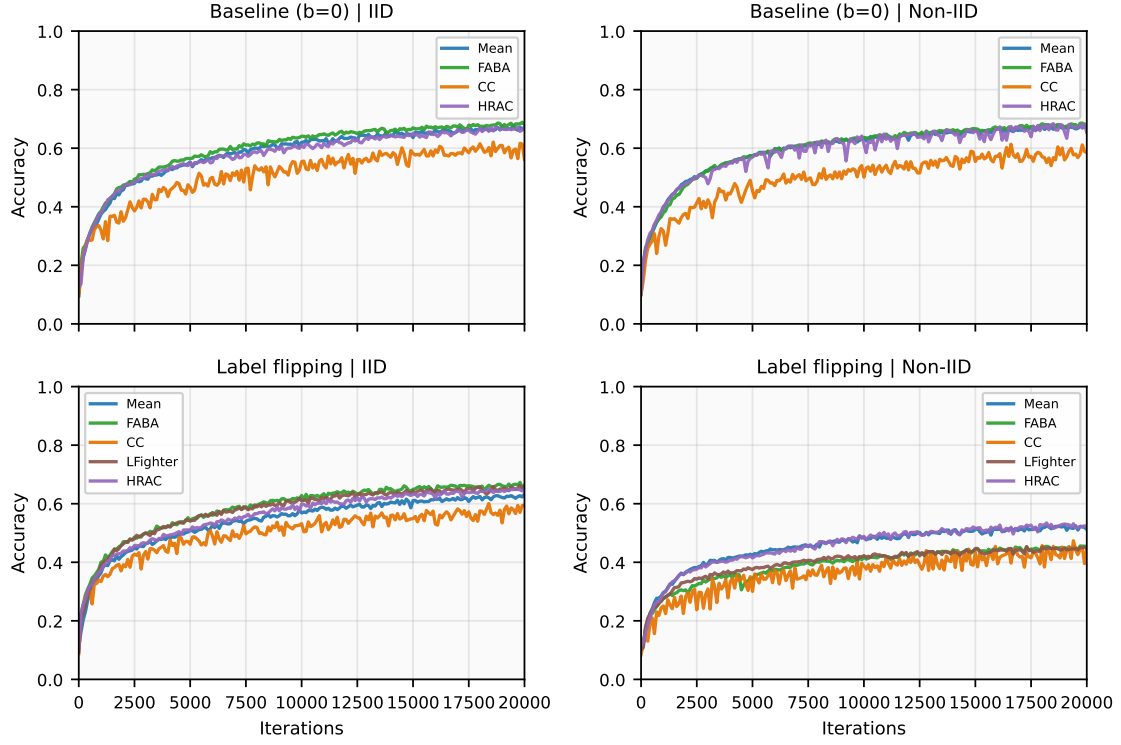


Figure 4.4: CIFAR-10 accuracy under benign training and label-flipping poisoning. The top row uses $b = 0$; the label-flipping row uses $b = 2$.

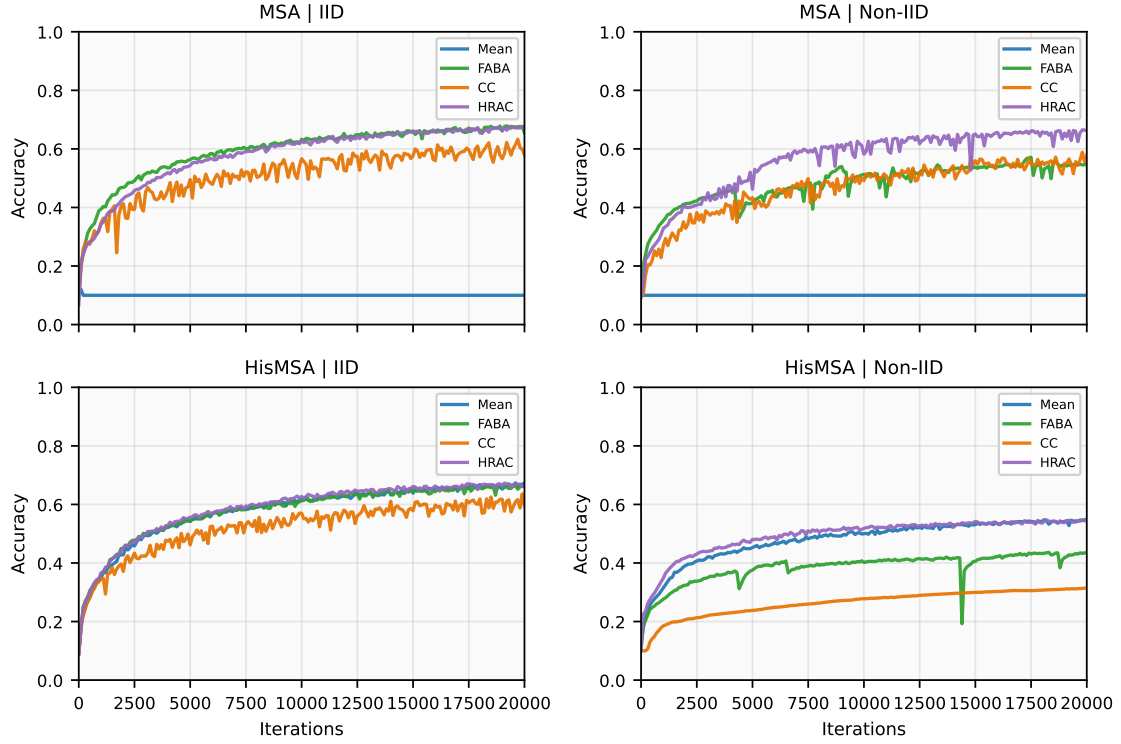


Figure 4.5: CIFAR-10 accuracy under MSA and HisMSA attacks with $b = 2$.

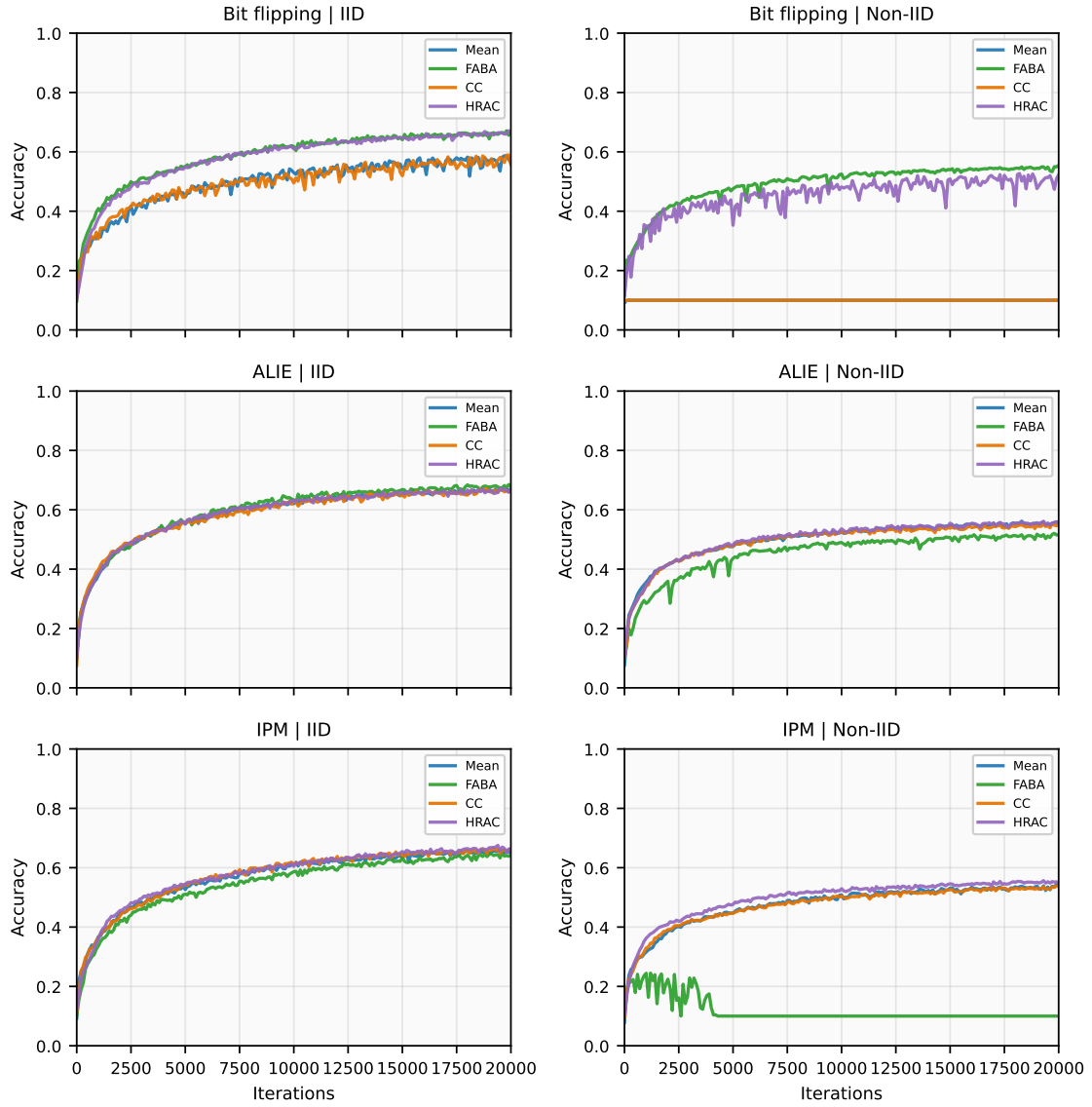


Figure 4.6: CIFAR-10 accuracy under bit-flipping, ALIE, and IPM attacks with $b = 2$.

Table 4.3: Final and best CIFAR-10 test accuracy extracted from cached logs for the main $b = 3$ attack setting. Entries are reported as final / best.

Setting	Partition	Mean	Best robust baseline	HRAC
Baseline	IID	0.6707 / 0.6780	FABA: 0.6808 / 0.6887	0.6698 / 0.6721
Baseline	Non-IID	0.6718 / 0.6770	FABA: 0.6820 / 0.6857	0.6540 / 0.6839
Label flipping	IID	0.5889 / 0.5951	FABA: 0.6697 / 0.6697	0.6113 / 0.6152
Label flipping	Non-IID	0.4391 / 0.4449	CC: 0.3713 / 0.3794	0.4405 / 0.4429
MSA	IID	0.4433 / 0.4464	FABA: 0.6630 / 0.6789	0.6618 / 0.6618
MSA	Non-IID	0.3369 / 0.3507	CC: 0.5479 / 0.5626	0.6466 / 0.6466
HisMSA	IID	0.6577 / 0.6593	FABA: 0.6478 / 0.6500	0.6493 / 0.6732
HisMSA	Non-IID	0.4734 / 0.4743	CC: 0.4863 / 0.4868	0.4874 / 0.4912
Bit flipping	IID	0.5006 / 0.5445	FABA: 0.6687 / 0.6729	0.6685 / 0.6719
Bit flipping	Non-IID	0.1000 / 0.1000	FABA: 0.4875 / 0.4885	0.4371 / 0.4604
ALIE	IID	0.6497 / 0.6758	CC: 0.6607 / 0.6753	0.6662 / 0.6718
ALIE	Non-IID	0.3585 / 0.3921	CC: 0.3695 / 0.3953	0.4805 / 0.4846
IPM	IID	0.6315 / 0.6348	CC: 0.6512 / 0.6546	0.6459 / 0.6476
IPM	Non-IID	0.4646 / 0.4733	CC: 0.4736 / 0.4740	0.4839 / 0.4931

Table 4.4: Final and best CIFAR-10 test accuracy extracted from cached logs for the supplementary $b = 2$ attack setting. Entries are reported as final / best.

Setting	Partition	Mean	Best robust baseline	HRAC
Label flipping	IID	0.6330 / 0.6330	FABA: 0.6638 / 0.6728	0.6494 / 0.6506
Label flipping	Non-IID	0.5118 / 0.5264	CC: 0.4461 / 0.4736	0.5215 / 0.5343
MSA	IID	0.1000 / 0.1201	FABA: 0.6544 / 0.6788	0.6785 / 0.6785
MSA	Non-IID	0.1000 / 0.1000	CC: 0.5780 / 0.5890	0.6628 / 0.6652
HisMSA	IID	0.6725 / 0.6738	FABA: 0.6615 / 0.6646	0.6629 / 0.6729
HisMSA	Non-IID	0.5498 / 0.5498	FABA: 0.4359 / 0.4363	0.5447 / 0.5472
Bit flipping	IID	0.5640 / 0.5869	FABA: 0.6642 / 0.6710	0.6665 / 0.6713
Bit flipping	Non-IID	0.1000 / 0.1000	FABA: 0.5544 / 0.5544	0.5227 / 0.5272
ALIE	IID	0.6711 / 0.6711	FABA: 0.6812 / 0.6852	0.6721 / 0.6721
ALIE	Non-IID	0.5584 / 0.5618	CC: 0.5522 / 0.5522	0.5571 / 0.5616
IPM	IID	0.6601 / 0.6601	CC: 0.6559 / 0.6667	0.6627 / 0.6753
IPM	Non-IID	0.5443 / 0.5443	CC: 0.5361 / 0.5388	0.5504 / 0.5565

the observed behaviour comes from one clipping rule or from the combination of scale control, client-specific history, and temporal weighting.

The ALIE ablation in Figure 4.7 shows that the global cap and residual clipping are not the decisive components in this run. Both corresponding variants stay close to full HRAC (Table 4.5). In contrast, removing ν weighting causes a large drop, and the global-cap-only variant performs similarly poorly. This suggests that the residual-change statistic is the main useful signal against ALIE in this experiment.

The MSA ablation in Figure 4.8 gives a different pattern. Removing the global cap reduces accuracy to near chance level, while removing residual clipping leaves a clear gap from full HRAC (Table 4.6). This indicates that MSA is controlled mainly by explicit scale limitation, with residual clipping adding a second history-aware constraint. The MSA ablation values come from a separate controlled ablation cache rather than the main comparison cache used in Table 4.3, so the full-HRAC reference value is not identical to the MSA entry in the main summary table.

Overall, the ablations show that HRAC’s components matter in different ways for different attacks. Temporal weighting is most useful for ALIE, while the global cap is essential for MSA and residual clipping gives additional protection. This supports keeping the three mechanisms together

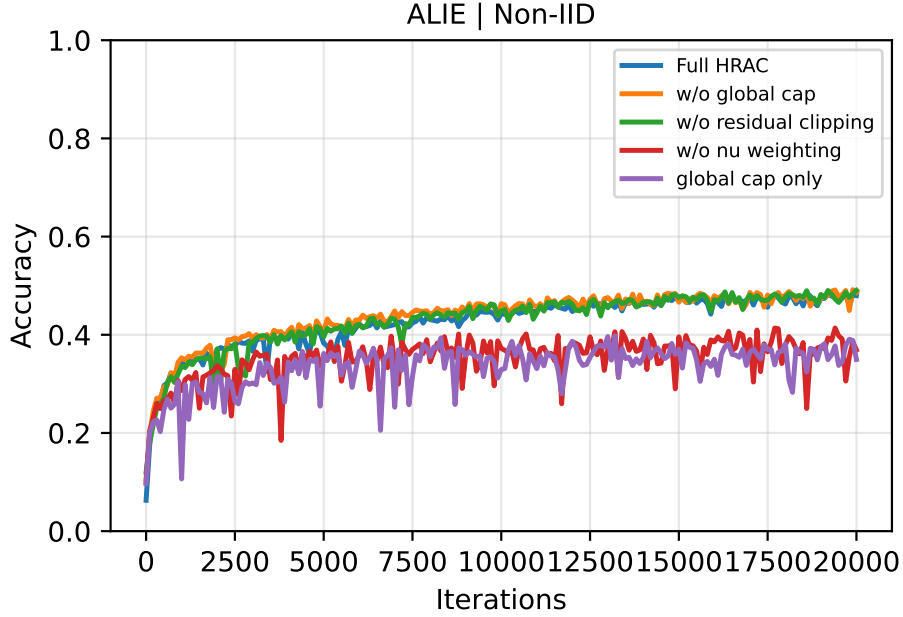


Figure 4.7: HRAC component ablation under ALIE on non-IID CIFAR-10 with $b = 3$.

Table 4.5: Final and best test accuracy for the ALIE non-IID ablation study.

Variant	Final	Best	Main effect
Full HRAC	0.4799	0.4843	Reference method
No global cap	0.4863	0.4919	Similar in ALIE
No residual clipping	0.4903	0.4903	Similar in ALIE
No ν weighting	0.3691	0.4139	Large degradation
Global cap only	0.3498	0.3979	Large degradation

rather than treating HRAC as a single clipping heuristic.

4.3.3 Diagnostic Analysis

The HRAC logs also allow the accuracy results to be connected to the internal statistics used by HRAC. During training, the implementation records per-round residual thresholds τ_i , residual scale statistics μ_i , residual-change statistics ν_i , normalised client weights, pre- and post-clipping norms, and bottom-weight summaries for controlled simulation runs.

The diagnostic plots are treated as supporting analysis rather than as the primary evidence for HRAC. The left panel tracks the average residual-change statistic ν_i for benign and Byzantine clients, the middle panel shows their normalised aggregation weights, and the right panel counts how many Byzantine clients appear among the three lowest-weighted clients. The main empirical claims in this chapter therefore rest on the accuracy curves, final/best accuracy summaries, and component ablations, while diagnostics are used to interpret those results.

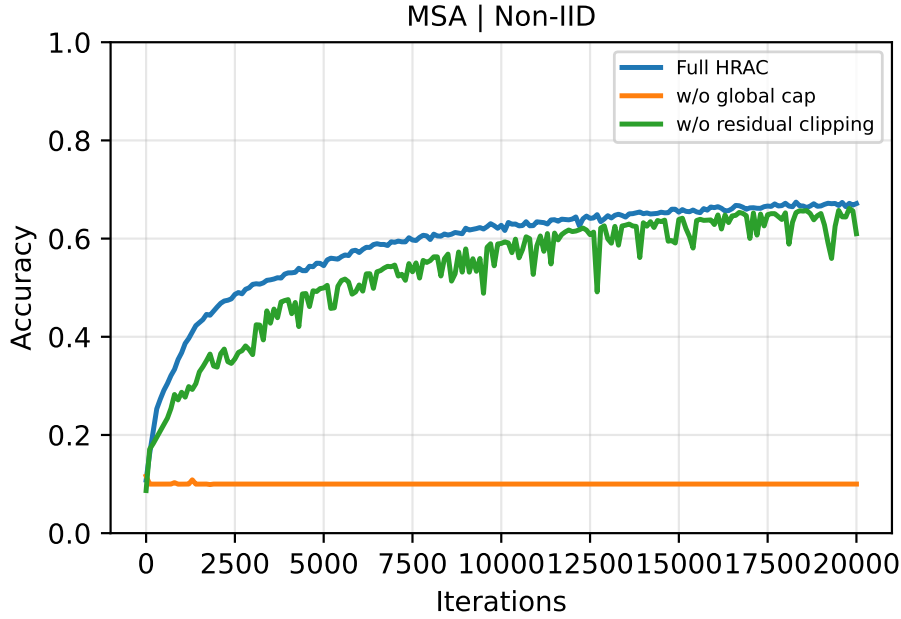


Figure 4.8: HRAC component ablation under MSA on non-IID CIFAR-10 with $b = 3$.

Table 4.6: Final and best test accuracy for the MSA non-IID ablation study. These controlled ablation runs use a separate cache from the main comparison in Table 4.3; therefore the full-HRAC reference value differs from the main-table HRAC value.

Variant	Final	Best
Full HRAC	0.6713	0.6743
No global cap	0.1000	0.1163
No residual clipping	0.6099	0.6615

Across most attacks, these diagnostics show the intended behaviour. Once the weighting stage becomes active, Byzantine clients tend to have larger residual-change statistics, receive lower average weights, and appear frequently in the bottom-weight set. This pattern is visible for bit flipping, ALIE, IPM, HisMSA, and IID label flipping. It suggests that the weighting rule is often suppressing the clients that generate abnormal residual dynamics.

The weaker cases are also informative. For MSA, strong accuracy does not always come with clean attacker identification in the weight diagnostics. This is consistent with the ablation result, where the global cap is the decisive component and limits the effect of shuffled or amplified updates even when the weights are not a reliable detector. For label flipping, the IID case still shows clear attacker down-weighting, but the non-IID case is harder. Benign label-skew already creates client-specific residual variation, so poisoned and benign ν_i trajectories overlap more strongly and the bottom-weight summary captures the attackers less consistently. This is why protocol-compliant label poisoning becomes more difficult when it is combined with heterogeneous data.

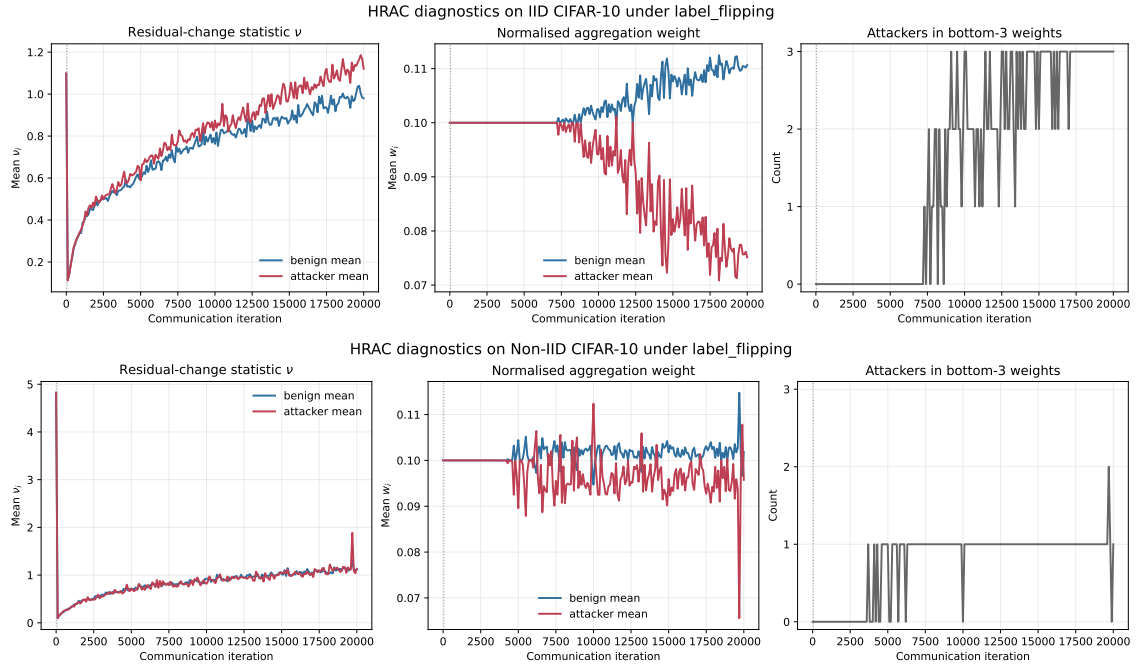


Figure 4.9: HRAC diagnostics under label-flipping attacks for IID and non-IID CIFAR-10.

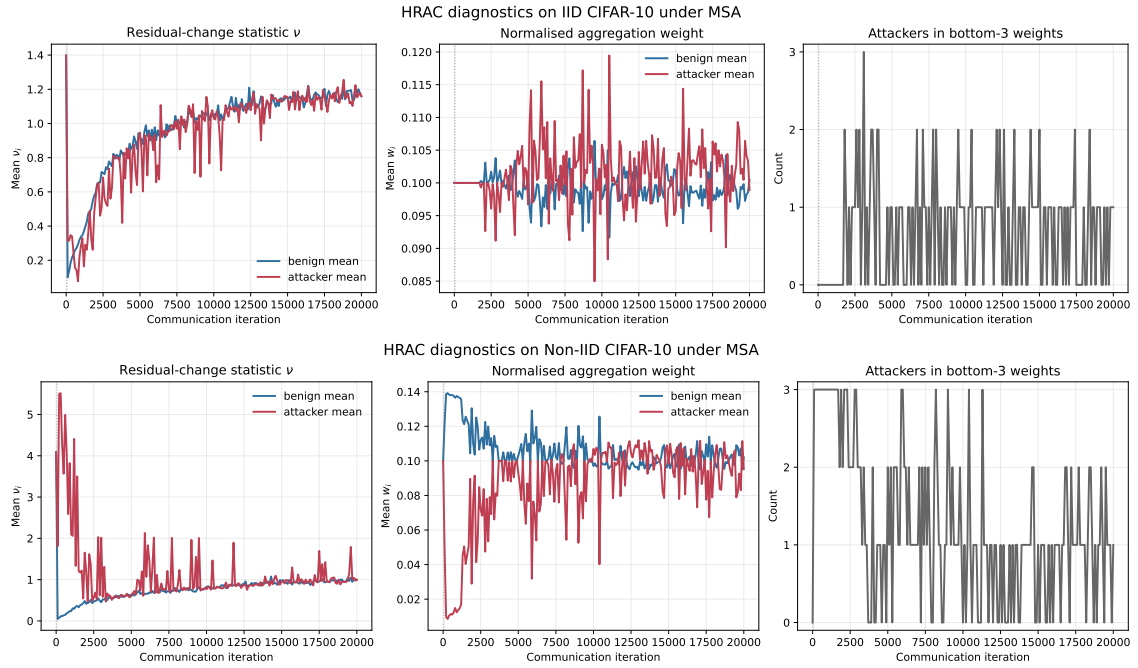


Figure 4.10: HRAC diagnostics under MSA attacks for IID and non-IID CIFAR-10.

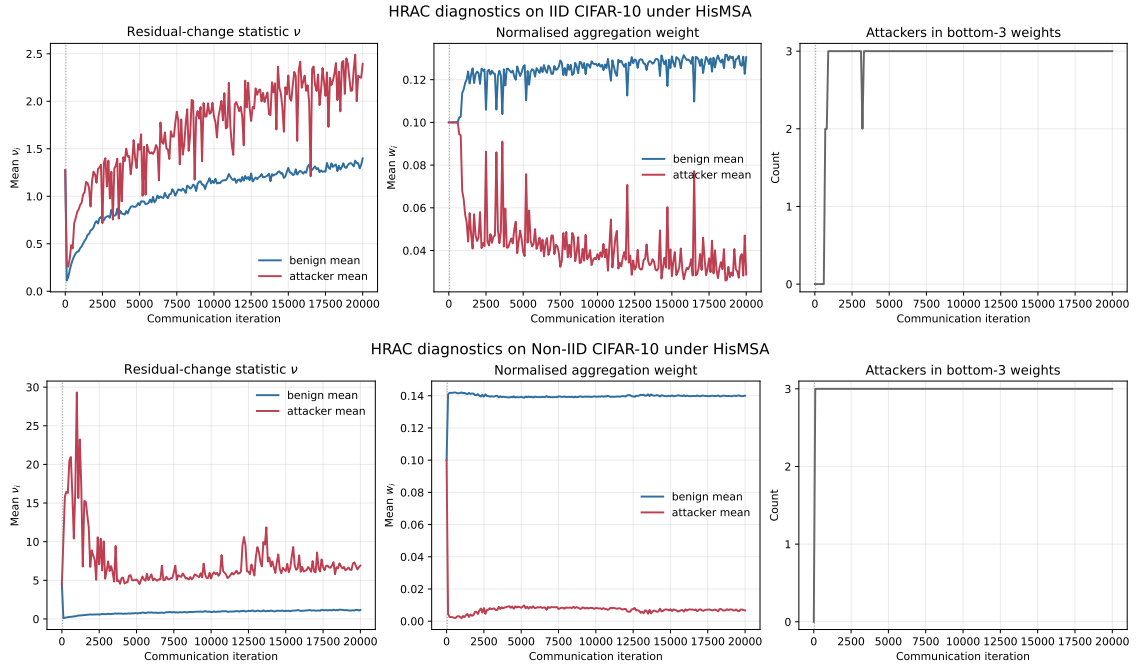


Figure 4.11: HRAC diagnostics under HisMSA attacks for IID and non-IID CIFAR-10.

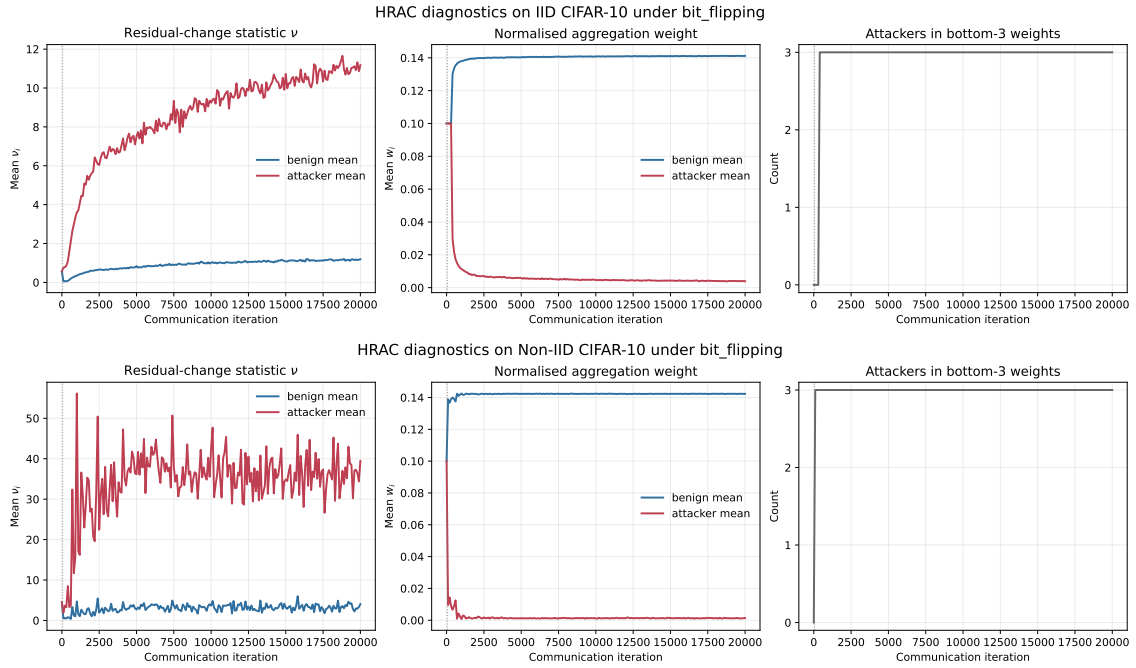


Figure 4.12: HRAC diagnostics under bit-flipping attacks for IID and non-IID CIFAR-10.

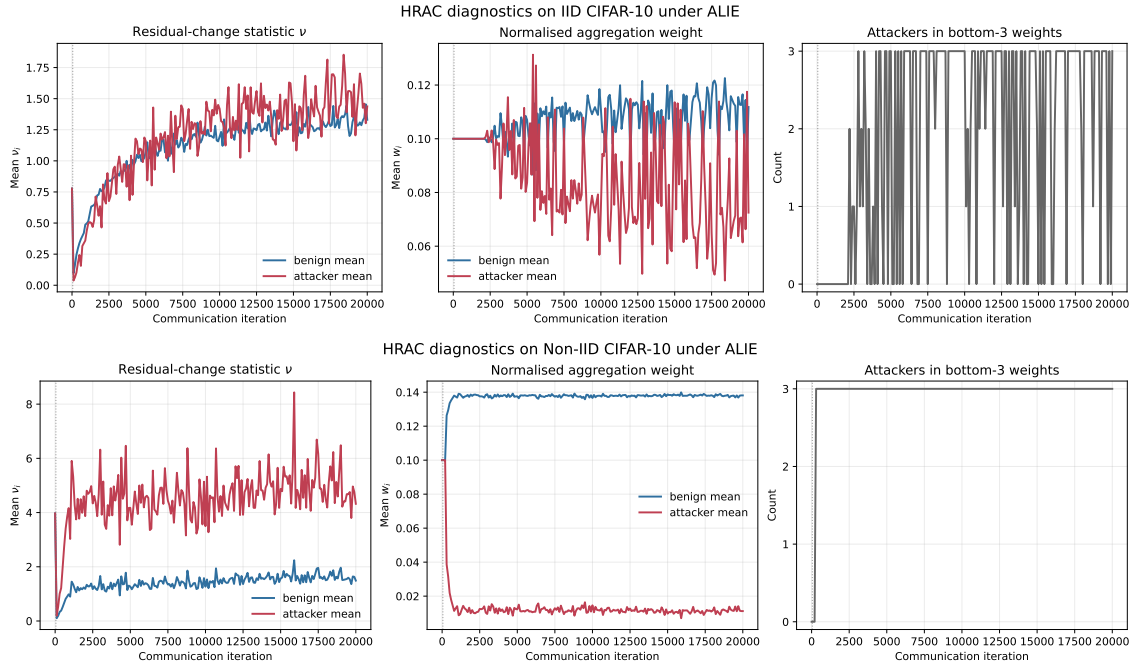


Figure 4.13: HRAC diagnostics under ALIE attacks for IID and non-IID CIFAR-10.

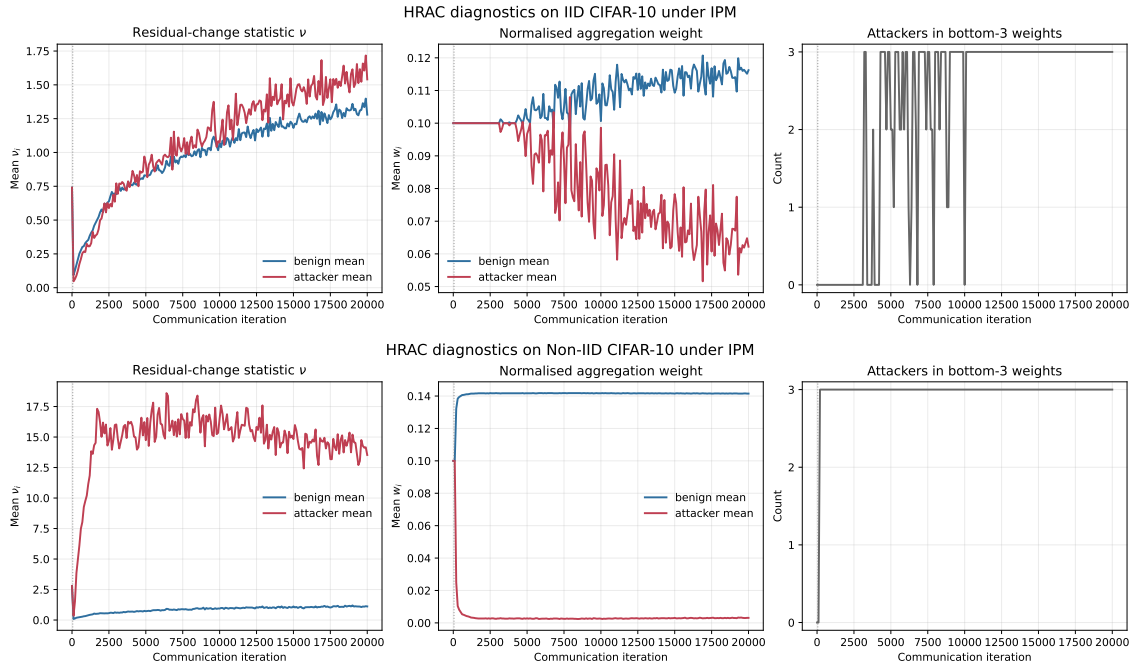


Figure 4.14: HRAC diagnostics under IPM attacks for IID and non-IID CIFAR-10.

4.4 Discussion and Limitations

The experimental results match the trade-off stated in the abstract. Mean aggregation can remain competitive for protocol-compliant poisoning under heterogeneous data, but it is not reliable under stronger model-poisoning attacks. Some robust baselines are strong in particular cases, but their relative ranking changes with the attack and partition. HRAC is most useful when malicious behaviour appears as structured or persistent deviation, as in MSA, HisMSA, non-IID ALIE, and non-IID IPM.

The mixed label-flipping and bit-flipping results are therefore part of the claim rather than an exception to it. Under IID label flipping, HRAC performs well because poisoned clients stand out more clearly from a relatively aligned benign population. The diagnostic plot supports this because attacker weights decline over training, and the bottom-weight set increasingly contains the Byzantine clients. Under non-IID label flipping, the difficulty comes from the interaction between the attack and the data partition rather than from label flipping alone. Benign label-skew already produces client-specific residual variation, so the ν trajectories of benign and poisoned clients overlap much more, and the bottom-weight summary captures the attackers less consistently. LFighter follows a similar pattern in the available label-flipping runs, where it is strong under IID data but weaker under the non-IID partition. Bit flipping is different: in the diagnostic plots, the attacker ν values separate strongly from the benign clients, the attacker weights are pushed close to zero, and the bottom-weight summary consistently contains the Byzantine clients. HRAC should therefore be read as a balanced robust aggregation strategy, not as a specialised label-poisoning defence.

The main limitations are the fixed HRAC hyperparameter setting and the restricted experiment grid. The current cached results use CIFAR-10, $N = 10$ clients, seed 100, $b = 2$ and $b = 3$ attack settings, and one label class per client for the non-IID partition. The attack parameters are also fixed in the reported runs, for example IPM uses $\epsilon = 0.1$ and MSA/HisMSA use shuffle probability 1.0 with scaling range $(0.1, 10.0)$. A stronger empirical study would rerun the same comparison across multiple seeds, datasets, client counts, attacker fractions, label-skew severities, and attack strengths.

Conclusions and future directions

This project studied Byzantine-robust aggregation under the trade-off between robustness and heterogeneity tolerance. Mean aggregation is not conservative enough for strong model-poisoning attacks, while robust aggregation can become too conservative when benign non-IID clients naturally deviate from the population.

HRAC addresses this trade-off by limiting the influence of unstable updates. It combines an EMA-smoothed median norm cap, per-client history-residual clipping, and delayed soft weighting based on residual-change statistics. The method compares each client mainly with its own historical behaviour rather than treating distance from a global centre as sufficient evidence of maliciousness.

On CIFAR-10, HRAC remains competitive in benign training in the reported cached runs. Its strongest improvements appear under MSA and under non-IID ALIE/IPM. The diagnostic plots also show strong attacker down-weighting in several settings, including bit flipping, although the final-accuracy tables show that FABAs can still outperform HRAC in some bit-flipping cases. The ablations support the component design: the global cap is essential under MSA, while ν -based temporal weighting is most important under ALIE.

The results also show the method’s limits. Label flipping remains more attack- and partition-dependent, especially when protocol-compliant poisoning is combined with non-IID data. The reported experiments are controlled cached runs on CIFAR-10 with $N = 10$, seed 100, $b = 2$ or $b = 3$, and one label class per client in the non-IID split, so they should be treated as single-seed trends rather than confidence-interval benchmarks. The next experimental step is not only to add more runs, but to vary the factors that could change the ranking: dataset and model family, client count, attacker fraction, label-skew severity, and attack strength. HRAC also assumes that the server can observe per-client updates and maintain per-client history statistics, so it is not directly compatible with standard secure aggregation without additional protocol support.



Convergence Analysis of HRAC

This appendix gives the conditional convergence argument supporting the brief statement in Chapter 3. The proof treats HRAC as a robust aggregation rule with bounded clipping distortion, bounded history-state growth, and bounded residual Byzantine weight mass, then inserts the resulting aggregation error into a standard smooth non-convex descent argument. It should be read as a bounded-influence stability analysis under stated assumptions, not as a proof that HRAC identifies Byzantine clients in all settings.

A.1 Introduction

The goal is to separate the proof into two parts. Rather than proving the code line by line, HRAC is abstracted as a robust aggregation rule and analysed as follows:

1. show that the HRAC aggregate remains close to the ideal honest conditional centre;
2. show that such an aggregation error can be absorbed into the descent analysis of smooth non-convex optimisation.

For the adaptive statistics, the main theorem-level target is not universal pointwise ordering between Byzantine and honest clients at every late-stage round. Instead, the primary claim is the weaker and empirically better-matched one: Byzantine clients exhibit a larger late-stage group-mean ν -band, which induces stronger penalty pressure and smaller weights once the activation threshold becomes active. In particular, the intended mechanism is not that honest statistics vanish

while Byzantine statistics remain nonzero. Rather, both groups may stay in positive late-stage bands, while Byzantine clients are typically elevated relative to the honest population. Pointwise separation is kept as a stronger sufficient refinement.

The clipping argument is stated at the level needed for this dissertation. To make the global cap compatible with the clipped-moment lemma of Sadiev et al. [38], HRAC uses a lagged history-EMA median scale as its global clipping radius. Thus the global threshold is $\mathcal{F}_t$ -measurable at round t , just like the residual threshold determined by the HRAC history state. The price of this proof route is a strong bounded honest-update assumption, so the result certifies bounded-influence behaviour rather than a full distribution-free FL guarantee.

A.2 Problem Setup

Let $x_t \in \mathbb{R}^d$ denote the global model at round t . Suppose there are N participating clients, partitioned into an honest set $\mathcal{H}$ and a Byzantine set $\mathcal{B}$, with

$$|\mathcal{H}| = H, \quad |\mathcal{B}| = B, \quad N = H + B.$$

At round t , the server receives client messages

$$\{\Delta_i^t\}_{i=1}^N \subset \mathbb{R}^d,$$

where Δ_i^t denotes a gradient-like descent direction, as in Chapters 2 and 3, and performs the global update

$$x_{t+1} = x_t - \eta_t g_t, \tag{A.1}$$

where g_t is the aggregate produced by HRAC.

The optimization proof below uses the conditional honest centre as the reference object, because the clipped-moment lemma of Sadiev et al. [38, Lemma 5.1] is stated conditionally on $\mathcal{F}_t$. For honest clients, define

$$m_i^t := \mathbb{E}[\Delta_i^t \mid \mathcal{F}_t], \quad i \in \mathcal{H}, \tag{A.2}$$

and the average honest conditional centre

$$\bar{m}_{\mathcal{H}}^t := \frac{1}{H} \sum_{i \in \mathcal{H}} m_i^t. \quad (\text{A.3})$$

The objective is to minimise a possibly non-convex function

$$f : \mathbb{R}^d \rightarrow \mathbb{R}.$$

A.3 HRAC Aggregation Rule

HRAC operates in three stages.

A.3.1 Global norm clipping

Define the current round median norm

$$M_t := \text{med}_{1 \leq j \leq N} \|\Delta_j^t\|. \quad (\text{A.4})$$

HRAC maintains a history-EMA global scale

$$s_{t+1} = \rho_g s_t + (1 - \rho_g) M_t, \quad 0 < \rho_g < 1, \quad (\text{A.5})$$

and uses the predictable global clipping cap with a fixed positive floor

$$B_t := c_g \max\{s_t, s_{\min}\}, \quad c_g \geq 1, \quad s_{\min} > 0. \quad (\text{A.6})$$

Thus $B_t \in \mathcal{F}_t$: the current round median M_t updates the next round's radius but is not used to clip the current round's updates. The globally clipped update is

$$\bar{\Delta}_i^t := \text{clip}(\Delta_i^t, B_t) := \begin{cases} \Delta_i^t, & \|\Delta_i^t\| \leq B_t, \\ B_t \frac{\Delta_i^t}{\|\Delta_i^t\|}, & \|\Delta_i^t\| > B_t. \end{cases} \quad (\text{A.7})$$

The implementation warm-starts the EMA scale from the first observed median and then applies (A.5). In the analysis below, the counted rounds start after this warm-start state is available, so the global cap used inside the proof is predictable. Equivalently, one may view the first current-median cap as a finite calibration prefix, which has no effect on the asymptotic neighbourhood.

A.3.2 Per-client history-residual clipping

For each client i , HRAC maintains a history state

$$(b_i^t, \mu_i^t, \nu_i^t),$$

where b_i^t is a running centre, μ_i^t is a scale estimate, and ν_i^t is a change statistic.

Define the residual

$$r_i^t := \bar{\Delta}_i^t - b_i^t, \tag{A.8}$$

and the adaptive residual clipping threshold

$$\tau_i^t := c\mu_i^t. \tag{A.9}$$

The clipped residual is

$$\hat{r}_i^t := \text{clip}(r_i^t, \tau_i^t) = \begin{cases} r_i^t, & \|r_i^t\| \leq \tau_i^t, \\ \tau_i^t \frac{r_i^t}{\|r_i^t\|}, & \|r_i^t\| > \tau_i^t. \end{cases} \tag{A.10}$$

The post-processed message becomes

$$\tilde{\Delta}_i^t := b_i^t + \hat{r}_i^t. \tag{A.11}$$

The history centre b_i^t follows the implemented EMA dynamics: for a newly appearing client, b_i^t is initialized by $\bar{\Delta}_i^t$; for later rounds,

$$b_i^{t+1} = \rho_b b_i^t + (1 - \rho_b) \Delta_{i,\text{safe}}^t, \tag{A.12}$$

where $\Delta_{i,\text{safe}}^t$ is the aggregation-path message used in the server history update. The reported

experiments use the reconstructed message $\tilde{\Delta}_i^t$ for this update:

$$\Delta_{i,\text{safe}}^t = \tilde{\Delta}_i^t. \quad (\text{A.13})$$

The uniform bound on this reconstructed message is proved below from the global cap and the convexity of the residual reconstruction step.

The residual scale follows the implemented EMA recursion

$$\mu_i^{t+1} = \rho_\mu \mu_i^t + (1 - \rho_\mu) u_i^t, \quad 0 < \rho_\mu < 1, \quad (\text{A.14})$$

where u_i^t is the residual-scale input produced by the globally capped aggregation/statistics path. The implementation initializes new clients with $\mu_i^t \leq 2B_t$; once $B_t \leq B_{\max}$, the capped-state geometry used below gives $0 \leq u_i^t \leq 2B_{\max}$. Thus the scale update is a code-level invariant rather than a distributional assumption.

A.3.3 Adaptive weighting

HRAC assigns weights $w_i^t \geq 0$ satisfying

$$\sum_{i=1}^N w_i^t = 1, \quad (\text{A.15})$$

where the weights are determined from the statistics $\{\nu_i^t\}_{i=1}^N$. For the conditional analysis below, w_i^t is computed from the history statistics available at the start of round t ; the current round updates only affect the next statistics state.

The final aggregate is

$$g_t = \sum_{i=1}^N w_i^t \tilde{\Delta}_i^t. \quad (\text{A.16})$$

A.4 Assumptions

Assumption 1 (Lower bounded objective). *f is bounded below: $f(x) \geq f^*$ for all $x \in \mathbb{R}^d$.*

Assumption 2 (L -smoothness). *$\|\nabla f(x) - \nabla f(y)\| \leq L \|x - y\|$ for all $x, y \in \mathbb{R}^d$.*

Assumption 3 (Bias of the honest conditional centre). *Conditioned on $\mathcal{F}_t$, the honest conditional*

centre has bounded bias:

$$\|\bar{m}_{\mathcal{H}}^t - \nabla f(x_t)\| \leq \xi, \quad (\text{A.17})$$

Assumption 4 (Bounded honest update magnitude). *There exists $G_H < \infty$ such that for all honest clients $i \in \mathcal{H}$ and all rounds t ,*

$$\|\Delta_i^t\| \leq G_H. \quad (\text{A.18})$$

Assumption 5 (Honest majority for robust median cap). *The Byzantine population is strictly less than the honest population:*

$$B < H \quad (\text{equivalently } B < N/2). \quad (\text{A.19})$$

Assumption 6 (Honest global clipping admissibility). *There exist constants $\alpha_g \in (1, 2]$ and $\sigma_{\text{gc}} \geq 0$ such that, for every honest client $i \in \mathcal{H}$ and every round t ,*

$$\|m_i^t\| \leq \frac{B_t}{2}, \quad (\text{A.20})$$

and

$$\mathbb{E}[\|\Delta_i^t - m_i^t\|^{\alpha_g} \mid \mathcal{F}_t] \leq \sigma_{\text{gc}}^{\alpha_g}. \quad (\text{A.21})$$

With $B_{\max} := c_g G_H$, we define

$$\mathcal{C}_{\text{gc}} := 18 B_{\max}^{2-\alpha_g} \sigma_{\text{gc}}^{\alpha_g}. \quad (\text{A.22})$$

Assumption 7 (Honest residual clipping admissibility). *For each honest client, define the residual conditional centre*

$$s_i^t := \mathbb{E}[r_i^t \mid \mathcal{F}_t]. \quad (\text{A.23})$$

There exist constants $\alpha_r \in (1, 2]$, $\sigma_{\text{rc}} \geq 0$, and $\kappa_\mu < \infty$ such that, for every honest client $i \in \mathcal{H}$ and every round t ,

$$\mathbb{E}[\|r_i^t\| \mid \mathcal{F}_t] \leq \kappa_\mu \mu_i^t, \quad c \geq 2\kappa_\mu, \quad (\text{A.24})$$

and

$$\mathbb{E}[\|r_i^t - s_i^t\|^{\alpha_r} \mid \mathcal{F}_t] \leq \sigma_{\text{rc}}^{\alpha_r}. \quad (\text{A.25})$$

With $B_{\max} := c_g G_H$ and $\tau_{\max} := 2c B_{\max}$, define

$$\mathcal{C}_{\text{rc}} := 18 \tau_{\max}^{2-\alpha_r} \sigma_{\text{rc}}^{\alpha_r}. \quad (\text{A.26})$$

Remark 1 (Admissibility and log-checkable scale certificates). The half-radius conditions are explicit admissibility assumptions. They are not automatic consequences of a current-round median; the point of the history-EMA global scale is to make B_t predictable, so that the clipped moment lemma of Sadiev et al. [38, Lemma 5.1] applies conditionally on $\mathcal{F}_t$. A sufficient global certificate is

$$M_m^t := \max_{i \in \mathcal{H}} \|m_i^t\| \leq \kappa_g \max\{s_t, s_{\min}\}, \quad c_g \geq 2\kappa_g.$$

For the residual layer, (A.24) says that μ_i^t does not underestimate the honest residual scale by more than κ_μ , with $c \geq 2\kappa_\mu$ providing the corresponding half-radius margin. Empirically, these requirements can be checked by logging moving-window ratios between honest update or residual scale proxies and the EMA scales s_t and μ_i^t .

Assumption 8 (Uniform Byzantine mass and honest-weight distortion control). *Define the residual Byzantine mass and the intra-honest weight distortion by*

$$\beta_t := \sum_{j \in \mathcal{B}} w_j^t, \quad \varepsilon_{w,t} := \sum_{i \in \mathcal{H}} \left| w_i^t - \frac{1}{H} \right|. \quad (\text{A.27})$$

There exist constants $\beta_\infty \in [0, 1]$ and $\varepsilon_{w,\infty} \geq 0$ such that the weights w_i^t are $\mathcal{F}_t$ -measurable and, for every round t ,

$$\beta_t \leq \beta_\infty, \quad (\text{A.28})$$

$$\varepsilon_{w,t} \leq \varepsilon_{w,\infty}, \quad (\text{A.29})$$

This assumption is not derived automatically from small Byzantine mass. Even with $\beta_t = 0$, heterogeneous honest clients can receive nonuniform weights because their ν_i histories differ. The theorem therefore keeps honest weight distortion as its own error-floor term.

Remark 2 (How to instantiate β_∞). Assumption 8 is stated abstractly so that the optimisation theorem stays compatible with the implemented HRAC weighting rule. In an idealized exponential-weight model,

$$w_i^t = \frac{\exp(-\lambda \nu_i^t)}{\sum_{k=1}^N \exp(-\lambda \nu_k^t)},$$

if the late-stage separation condition $\nu_j^t \geq \nu_i^t + \gamma$ holds for all $j \in \mathcal{B}$, $i \in \mathcal{H}$, then one may take

$$\beta_\infty := \frac{B e^{-\lambda \gamma}}{H + B e^{-\lambda \gamma}}. \quad (\text{A.30})$$

Thus the Byzantine weight mass becomes a fully explicit closed constant. The remaining Byzantine

energy factor is supplied pathwise by Lemma 6, which gives the processed-message bound $\|\tilde{\Delta}_j^t\| \leq B_{\max} = c_g G_H$.

Lemma 1 (Sadiev clipped moment lemma). *Let $X \in \mathbb{R}^d$ be a random vector, let $x = \mathbb{E}[X \mid \mathcal{F}]$, and define $\hat{X} = \text{clip}(X, \lambda)$. Following the clipped moment lemma of Sadiev et al. [38, Lemma 5.1], if*

$$\mathbb{E}[\|X - x\|^\alpha \mid \mathcal{F}] \leq \sigma^\alpha, \quad \|x\| \leq \frac{\lambda}{2}, \quad \alpha \in (1, 2],$$

then

$$\|\mathbb{E}[\hat{X} \mid \mathcal{F}] - x\| \leq \frac{2^\alpha \sigma^\alpha}{\lambda^{\alpha-1}}, \quad (\text{A.31})$$

$$\mathbb{E}[\|\hat{X} - x\|^2 \mid \mathcal{F}] \leq 18\lambda^{2-\alpha}\sigma^\alpha, \quad (\text{A.32})$$

$$\mathbb{E}[\|\hat{X} - \mathbb{E}[\hat{X} \mid \mathcal{F}]\|^2 \mid \mathcal{F}] \leq 18\lambda^{2-\alpha}\sigma^\alpha. \quad (\text{A.33})$$

Proof. This is Lemma 5.1 of Sadiev et al. [38], specialized here to the notation used by HRAC. The same result is proved in their Appendix C as Lemma C.1. $\square$

Lemma 2 (Protected-centre clipping distortion). *Let $x \in \mathbb{R}^d$, $X \in \mathbb{R}^d$, and $\lambda > 0$. If $\|x\| \leq \lambda$, then*

$$\|\text{clip}(X, \lambda) - X\| = (\|X\| - \lambda)_+ \leq \|X - x\|. \quad (\text{A.34})$$

Proof. If $\|X\| \leq \lambda$, both sides give a zero clipping distortion. If $\|X\| > \lambda$, then

$$\|\text{clip}(X, \lambda) - X\| = \|X\| - \lambda \leq \|X\| - \|x\| \leq \|X - x\|,$$

where the last step is the reverse triangle inequality. $\square$

Proposition 1 (Pointwise clipped-message control from Sadiev admissibility). *Assume that the weights w_i^t used at round t are predictable ($\mathcal{F}_t$ -measurable). Suppose Assumptions 4, 5, 6, and 7 hold, and HRAC uses the history-state update rules (A.12)–(A.14). Then Lemma 1, i.e., Sadiev et al. [38, Lemma 5.1], gives the global clipped-estimator control*

$$\mathbb{E}[\|\bar{\Delta}_i^t - m_i^t\|^2 \mid \mathcal{F}_t] \leq C_{\text{gc}}, \quad (\text{A.35})$$

for every honest client $i \in \mathcal{H}$. Moreover, the scale-tracking condition (A.24) implies the residual Sadiev admissibility condition

$$\|s_i^t\| \leq \frac{\tau_i^t}{2}, \quad (\text{A.36})$$

and another application of Sadiev et al. [38, Lemma 5.1] gives the residual clipped-estimator control

$$\mathbb{E} \left[\left\| \hat{r}_i^t - s_i^t \right\|^2 \mid \mathcal{F}_t \right] \leq \mathcal{C}_{\text{rc}}. \quad (\text{A.37})$$

Consequently, the two-stage honest clipped message satisfies

$$\mathbb{E} \left[\sum_{i \in \mathcal{H}} w_i^t \left\| \tilde{\Delta}_i^t - m_i^t \right\|^2 \mid \mathcal{F}_t \right] \leq 2\mathcal{C}_{\text{gc}} + 2\mathcal{C}_{\text{rc}}. \quad (\text{A.38})$$

Proof. For the global clipping layer, $B_t = c_g \max\{s_t, s_{\min}\}$ is $\mathcal{F}_t$ -measurable by (A.5)–(A.6). Applying Lemma 1, the clipped moment lemma of Sadiev et al. [38, Lemma 5.1], with

$$(X, x, \lambda, \alpha) = (\Delta_i^t, m_i^t, B_t, \alpha_g)$$

and using Assumption 6 gives

$$\mathbb{E} \left[\left\| \bar{\Delta}_i^t - m_i^t \right\|^2 \mid \mathcal{F}_t \right] \leq 18B_t^{2-\alpha_g} \sigma_{\text{gc}}^{\alpha_g} \leq 18B_{\max}^{2-\alpha_g} \sigma_{\text{gc}}^{\alpha_g} = \mathcal{C}_{\text{gc}}.$$

where the second inequality uses Corollary 1. Because w_i^t is $\mathcal{F}_t$ -measurable and $\sum_{i \in \mathcal{H}} w_i^t \leq 1$,

$$\mathbb{E} \left[\sum_{i \in \mathcal{H}} w_i^t \left\| \bar{\Delta}_i^t - m_i^t \right\|^2 \right] \leq \mathcal{C}_{\text{gc}}.$$

This is the per-round global clipped-message control used below.

For the residual layer, Jensen's inequality and (A.24) give

$$\|s_i^t\| = \|\mathbb{E}[r_i^t \mid \mathcal{F}_t]\| \leq \mathbb{E}[\|r_i^t\| \mid \mathcal{F}_t] \leq \kappa_\mu \mu_i^t \leq \frac{c\mu_i^t}{2} = \frac{\tau_i^t}{2},$$

which proves (A.36). Since $\tau_i^t = c\mu_i^t$ is determined by the history state, it is $\mathcal{F}_t$ -measurable.

Applying the clipped moment lemma of Sadiev et al. [38, Lemma 5.1], Lemma 1, with

$$(X, x, \lambda, \alpha) = (r_i^t, s_i^t, \tau_i^t, \alpha_r)$$

and using (A.25) gives

$$\mathbb{E} \left[\left\| \hat{r}_i^t - s_i^t \right\|^2 \mid \mathcal{F}_t \right] \leq 18(\tau_i^t)^{2-\alpha_r} \sigma_{\text{rc}}^{\alpha_r} \leq 18\tau_{\max}^{2-\alpha_r} \sigma_{\text{rc}}^{\alpha_r} = \mathcal{C}_{\text{rc}},$$

where the second inequality uses Lemma 6.

It remains to connect the two clipped-estimator controls to the actual HRAC message. Let $\bar{m}_i^t := \mathbb{E}[\bar{\Delta}_i^t \mid \mathcal{F}_t]$. Since b_i^t is $\mathcal{F}_t$ -measurable and $r_i^t = \bar{\Delta}_i^t - b_i^t$, we have

$$b_i^t + s_i^t = b_i^t + \mathbb{E}[r_i^t \mid \mathcal{F}_t] = \mathbb{E}[\bar{\Delta}_i^t \mid \mathcal{F}_t] = \bar{m}_i^t.$$

Therefore

$$\tilde{\Delta}_i^t - m_i^t = (\hat{r}_i^t - s_i^t) + (\bar{m}_i^t - m_i^t).$$

Using $\|a + b\|^2 \leq 2\|a\|^2 + 2\|b\|^2$, Jensen's inequality, and (A.35),

$$\|\bar{m}_i^t - m_i^t\|^2 = \|\mathbb{E}[\bar{\Delta}_i^t - m_i^t \mid \mathcal{F}_t]\|^2 \leq \mathbb{E}[\|\bar{\Delta}_i^t - m_i^t\|^2 \mid \mathcal{F}_t] \leq \mathcal{C}_{\text{gc}}.$$

Combining this bound with (A.37), multiplying by the $\mathcal{F}_t$ -measurable weights, and using $\sum_{i \in \mathcal{H}} w_i^t \leq 1$ proves (A.38). Thus both the global clipping layer and the residual clipping layer are controlled by the clipped-moment lemma of Sadiev et al. [38, Lemma 5.1]; the residual admissibility is supplied by the large- c scale-tracking condition. $\square$

A.5 Basic Properties of HRAC

Lemma 3 (Boundedness of globally clipped inputs). *For every round t and every client i ,*

$$\|\bar{\Delta}_i^t\| \leq B_t, \tag{A.39}$$

Proof. The bound $\|\bar{\Delta}_i^t\| \leq B_t$ follows directly from the definition of global clipping in (A.7). $\square$

Lemma 4 (EMA median-scale protection under honest majority). *Under Assumptions 5, 4, suppose the implementation warm-starts the global scale from a median observed under the same honest-majority condition, so $s_0 = M_{\text{init}}$, and choose the fixed floor with $0 < s_{\min} \leq G_H$. Then the round median M_t in (A.4) and the EMA scale s_t in (A.5) satisfy*

$$M_t \leq G_H, \quad s_t \leq G_H, \quad B_t = c_g \max\{s_t, s_{\min}\} \leq c_g G_H. \tag{A.40}$$

Proof. Because $H > B$, more than half of the N samples are honest, so every median order statistic is bracketed by honest samples. In particular, the median cannot exceed the maximum

honest norm. Assumption 4 therefore gives $M_t \leq G_H$ for every counted round. The same median argument applies to the warm-start median M_{init} , so $s_0 = M_{\text{init}} \leq G_H$. The EMA recursion then gives by induction

$$s_{t+1} = \rho_g s_t + (1 - \rho_g) M_t \leq \rho_g G_H + (1 - \rho_g) G_H = G_H.$$

Finally $s_{\min} \leq G_H$ by the fixed floor choice, so $\max\{s_t, s_{\min}\} \leq G_H$ and $B_t \leq c_g G_H$. $\square$

Corollary 1 (Uniform cap bound from honest majority and bounded honest updates). *Under Assumptions 5, 4, with the code-aligned warm-start and $0 < s_{\min} \leq G_H$,*

$$B_t \leq c_g G_H =: B_{\max}, \quad \forall t. \quad (\text{A.41})$$

Proof. This is the final inequality of Lemma 4. $\square$

Lemma 5 (EMA boundedness for μ_i^t). *Suppose nonnegative scalars s_t satisfy $s_t \leq S_{\max}$ for all t , and $\mu_{t+1} = \rho_\mu \mu_t + (1 - \rho_\mu) s_t$ with $0 < \rho_\mu < 1$ and $\mu_0 \leq S_{\max}$. Then $\mu_t \leq S_{\max}$ for all t .*

Proof. Induction with convexity:

$$\mu_{t+1} = \rho_\mu \mu_t + (1 - \rho_\mu) s_t \leq \rho_\mu S_{\max} + (1 - \rho_\mu) S_{\max} = S_{\max}.$$

$\square$

Lemma 6 (Bounded history states under bounded global cap). *Assume there exists $B_{\max} < \infty$ such that $B_t \leq B_{\max}$ for all rounds t . The reported implementation uses $\Delta_{i,\text{safe}}^t = \tilde{\Delta}_i^t$ in the history update. Under the HRAC history-state update rules (A.12)–(A.14), for every client i and every t :*

$$\|b_i^t\| \leq B_{\max}, \quad \|r_i^t\| \leq 2B_{\max}, \quad \mu_i^t \leq 2B_{\max}, \quad \tau_i^t = c\mu_i^t \leq 2cB_{\max}. \quad (\text{A.42})$$

Moreover, the post-processed message is deterministically bounded:

$$\|\tilde{\Delta}_i^t\| \leq B_{\max}. \quad (\text{A.43})$$

Proof. For b_i^t , the initialization satisfies $\|b_i^0\| \leq B_{\max}$ since it is set from a globally clipped message. Suppose $\|b_i^t\| \leq B_{\max}$. Because $\|\tilde{\Delta}_i^t\| \leq B_t \leq B_{\max}$, residual reconstruction has the form

$$\tilde{\Delta}_i^t = b_i^t + \hat{r}_i^t = (1 - \alpha_i^t) b_i^t + \alpha_i^t \bar{\Delta}_i^t, \quad 0 \leq \alpha_i^t \leq 1.$$

Both endpoints of this convex combination lie in the closed $B_{\max}$ -ball, so $\|\tilde{\Delta}_i^t\| \leq B_{\max}$. The implemented history update uses $\Delta_{i,\text{safe}}^t = \tilde{\Delta}_i^t$, hence

$$b_i^{t+1} = \rho_b b_i^t + (1 - \rho_b) \tilde{\Delta}_i^t.$$

Again by convexity, $\|b_i^{t+1}\| \leq B_{\max}$. Induction gives $\|b_i^t\| \leq B_{\max}$ and $\|\tilde{\Delta}_i^t\| \leq B_{\max}$ for all t .

Because $\|\bar{\Delta}_i^t\| \leq B_t \leq B_{\max}$,

$$\|r_i^t\| = \|\bar{\Delta}_i^t - b_i^t\| \leq \|\bar{\Delta}_i^t\| + \|b_i^t\| \leq 2B_{\max}.$$

The implemented initialization gives $\mu_i^t \leq 2B_t \leq 2B_{\max}$ when a client first appears. The scale input u_i^t in (A.14) is computed from residuals between globally capped messages and the bounded history state. Since both references are bounded by $B_{\max}$, the residual scale input satisfies $0 \leq u_i^t \leq 2B_{\max}$. Lemma 5 with $S_{\max} = 2B_{\max}$ gives $\mu_i^t \leq 2B_{\max}$, and hence $\tau_i^t = c\mu_i^t \leq 2cB_{\max}$. $\square$

Lemma 7 (Aggregate second moment from deterministic cap chain). *Under Assumptions 5 and 4, together with the HRAC history-state update rules (A.12)–(A.14), for every t , the aggregation output satisfies*

$$\mathbb{E}[\|g_t\|^2 \mid \mathcal{F}_t] \leq B_{\max}^2, \quad \text{hence} \quad \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|g_t\|^2 \leq B_{\max}^2. \quad (\text{A.44})$$

Proof. By Corollary 1, $B_t \leq B_{\max}$ for all t . Lemma 6 then gives $\|\tilde{\Delta}_i^t\| \leq B_{\max}$ deterministically for every client i . Write $g_t = \sum_{i=1}^N w_i^t \tilde{\Delta}_i^t$ with $\sum_i w_i^t = 1$, $w_i^t \geq 0$. Jensen's inequality yields:

$$\|g_t\|^2 = \left\| \sum_{i=1}^N w_i^t \tilde{\Delta}_i^t \right\|^2 \leq \sum_{i=1}^N w_i^t \|\tilde{\Delta}_i^t\|^2 \leq \sum_{i=1}^N w_i^t B_{\max}^2 = B_{\max}^2.$$

Taking expectations preserves the bound. $\square$

Lemma 8 (Residual clipping bias identity). *For every client i and round t ,*

$$\|\hat{r}_i^t - r_i^t\| = (\|r_i^t\| - \tau_i^t)_+, \quad (\text{A.45})$$

where $(a)_+ := \max\{a, 0\}$. As a consequence,

$$\|\tilde{\Delta}_i^t - \bar{\Delta}_i^t\| = (\|r_i^t\| - \tau_i^t)_+. \quad (\text{A.46})$$

Proof. If $\|r_i^t\| \leq \tau_i^t$, then $\hat{r}_i^t = r_i^t$, so the left-hand side is 0, equal to $(\|r_i^t\| - \tau_i^t)_+$.

If $\|r_i^t\| > \tau_i^t$, then

$$\hat{r}_i^t = r_i^t \cdot \frac{\tau_i^t}{\|r_i^t\|},$$

hence

$$\|\hat{r}_i^t - r_i^t\| = \left\| r_i^t \left(\frac{\tau_i^t}{\|r_i^t\|} - 1 \right) \right\| = \|r_i^t\| - \tau_i^t.$$

This proves (A.45). Since

$$\tilde{\Delta}_i^t - \bar{\Delta}_i^t = (b_i^t + \hat{r}_i^t) - (b_i^t + r_i^t) = \hat{r}_i^t - r_i^t,$$

we obtain (A.46). $\square$

A.6 Aggregation Error Analysis

Define the aggregation error

$$e_t := g_t - \bar{m}_{\mathcal{H}}^t. \quad (\text{A.47})$$

Lemma 9 (Aggregation error decomposition). *The error e_t admits the decomposition*

$$e_t = \underbrace{\sum_{i \in \mathcal{H}} w_i^t (\tilde{\Delta}_i^t - m_i^t)}_{=: A_t} + \underbrace{\sum_{i \in \mathcal{H}} \left(w_i^t - \frac{1}{H} \right) m_i^t}_{=: W_t} + \underbrace{\sum_{j \in \mathcal{B}} w_j^t \tilde{\Delta}_j^t}_{=: M_t}. \quad (\text{A.48})$$

Proof. From (A.16),

$$g_t = \sum_{i \in \mathcal{H}} w_i^t \tilde{\Delta}_i^t + \sum_{j \in \mathcal{B}} w_j^t \tilde{\Delta}_j^t.$$

Subtract $\bar{m}_{\mathcal{H}}^t = \frac{1}{H} \sum_{i \in \mathcal{H}} m_i^t$ and add/subtract $\sum_{i \in \mathcal{H}} w_i^t m_i^t$ to obtain

$$g_t - \bar{m}_{\mathcal{H}}^t = \sum_{i \in \mathcal{H}} w_i^t (\tilde{\Delta}_i^t - m_i^t) + \sum_{i \in \mathcal{H}} \left(w_i^t - \frac{1}{H} \right) m_i^t + \sum_{j \in \mathcal{B}} w_j^t \tilde{\Delta}_j^t.$$

$\square$

Lemma 10 (Bounding each term of the decomposition).

$$\|A_t\|^2 \leq \sum_{i \in \mathcal{H}} w_i^t \left\| \tilde{\Delta}_i^t - m_i^t \right\|^2, \quad (\text{A.49})$$

$$\|W_t\|^2 \leq \varepsilon_{w,t}^2 \cdot \max_{i \in \mathcal{H}} \|m_i^t\|^2 \leq \varepsilon_{w,t}^2 G_H^2, \quad (\text{A.50})$$

$$\|M_t\|^2 \leq \beta_t \sum_{j \in \mathcal{B}} w_j^t \left\| \tilde{\Delta}_j^t \right\|^2. \quad (\text{A.51})$$

Consequently,

$$\|e_t\|^2 \leq 3 \|A_t\|^2 + 3 \|W_t\|^2 + 3 \|M_t\|^2. \quad (\text{A.52})$$

Proof. For A_t , Jensen's inequality gives

$$\|A_t\|^2 = \left\| \sum_{i \in \mathcal{H}} w_i^t (\tilde{\Delta}_i^t - m_i^t) \right\|^2 \leq \sum_{i \in \mathcal{H}} w_i^t \left\| \tilde{\Delta}_i^t - m_i^t \right\|^2.$$

More explicitly, since $\sum_{i \in \mathcal{H}} w_i^t = 1 - \beta_t \leq 1$, weighted Cauchy–Schwarz yields

$$\|A_t\|^2 \leq \left(\sum_{i \in \mathcal{H}} w_i^t \right) \left(\sum_{i \in \mathcal{H}} w_i^t \left\| \tilde{\Delta}_i^t - m_i^t \right\|^2 \right) \leq \sum_{i \in \mathcal{H}} w_i^t \left\| \tilde{\Delta}_i^t - m_i^t \right\|^2.$$

For W_t ,

$$\|W_t\| \leq \sum_{i \in \mathcal{H}} \left| w_i^t - \frac{1}{H} \right| \|m_i^t\| \leq \varepsilon_{w,t} \max_{i \in \mathcal{H}} \|m_i^t\|.$$

Since $\|m_i^t\| \leq \mathbb{E}[\|\Delta_i^t\| \mid \mathcal{F}_t] \leq G_H$ by Assumption 4, this yields (A.50) after squaring.

For M_t , by Cauchy–Schwarz and $\sum_{j \in \mathcal{B}} w_j^t = \beta_t$,

$$\|M_t\| \leq \sum_{j \in \mathcal{B}} w_j^t \left\| \tilde{\Delta}_j^t \right\|$$

and therefore

$$\|M_t\|^2 \leq \left(\sum_{j \in \mathcal{B}} w_j^t \right) \left(\sum_{j \in \mathcal{B}} w_j^t \left\| \tilde{\Delta}_j^t \right\|^2 \right) = \beta_t \sum_{j \in \mathcal{B}} w_j^t \left\| \tilde{\Delta}_j^t \right\|^2.$$

which is exactly (A.51).

Finally,

$$e_t = A_t + W_t + M_t,$$

so

$$\|e_t\|^2 \leq 3 \|A_t\|^2 + 3 \|W_t\|^2 + 3 \|M_t\|^2.$$

□

A.7 From Honest Conditional Centre to the True Gradient

Lemma 11 (Deviation of the honest conditional centre from the true gradient). *Under Assumption 3,*

$$\|\bar{m}_{\mathcal{H}}^t - \nabla f(x_t)\|^2 \leq \xi^2. \quad (\text{A.53})$$

Proof. This is exactly Assumption 3, squared. □

A.8 One-Step Descent Analysis

Lemma 12 (One-step descent inequality). *Suppose Assumptions 1 to 5 hold. Then, for any step size $\eta_t > 0$,*

$$\begin{aligned} \mathbb{E}[f(x_{t+1})] &\leq \mathbb{E}[f(x_t)] - \frac{\eta_t}{2} \mathbb{E} \|\nabla f(x_t)\|^2 \\ &\quad + \eta_t \mathbb{E} \|e_t\|^2 + \eta_t \xi^2 + \frac{L\eta_t^2 - \eta_t}{2} \mathbb{E} \|g_t\|^2. \end{aligned} \quad (\text{A.54})$$

In particular, if $0 < \eta_t \leq 1/L$, then

$$\mathbb{E}[f(x_{t+1})] \leq \mathbb{E}[f(x_t)] - \frac{\eta_t}{2} \mathbb{E} \|\nabla f(x_t)\|^2 + \eta_t \mathbb{E} \|e_t\|^2 + \eta_t \xi^2. \quad (\text{A.55})$$

Proof. By Assumption 2,

$$f(x_{t+1}) \leq f(x_t) + \langle \nabla f(x_t), x_{t+1} - x_t \rangle + \frac{L}{2} \|x_{t+1} - x_t\|^2.$$

Substituting $x_{t+1} = x_t - \eta_t g_t$ gives

$$f(x_{t+1}) \leq f(x_t) - \eta_t \langle \nabla f(x_t), g_t \rangle + \frac{L\eta_t^2}{2} \|g_t\|^2.$$

Using the identity $-\langle a, b \rangle = -\frac{1}{2} \|a\|^2 - \frac{1}{2} \|b\|^2 + \frac{1}{2} \|a - b\|^2$, we can rewrite the inner product exactly as:

$$-\langle \nabla f(x_t), g_t \rangle = -\frac{1}{2} \|\nabla f(x_t)\|^2 - \frac{1}{2} \|g_t\|^2 + \frac{1}{2} \|g_t - \nabla f(x_t)\|^2.$$

Now, using the decomposition $g_t = \bar{m}_{\mathcal{H}}^t + e_t$, and applying the inequality $\|a + b\|^2 \leq 2\|a\|^2 + 2\|b\|^2$, we bound the distance between the aggregate and the true gradient:

$$\|g_t - \nabla f(x_t)\|^2 = \|e_t + \bar{m}_{\mathcal{H}}^t - \nabla f(x_t)\|^2 \leq 2\|e_t\|^2 + 2\|\bar{m}_{\mathcal{H}}^t - \nabla f(x_t)\|^2.$$

Substituting this back into the smoothness inequality yields:

$$f(x_{t+1}) \leq f(x_t) - \frac{\eta_t}{2} \|\nabla f(x_t)\|^2 + \frac{L\eta_t^2 - \eta_t}{2} \|g_t\|^2 + \eta_t \|e_t\|^2 + \eta_t \|\bar{m}_{\mathcal{H}}^t - \nabla f(x_t)\|^2.$$

Taking the conditional expectation $\mathbb{E}[\cdot \mid \mathcal{F}_t]$ on both sides and applying Lemma 11 for the honest centre deviation yields (A.54). If $0 < \eta_t \leq 1/L$, then $(L\eta_t^2 - \eta_t)/2 \leq 0$, so dropping the nonpositive term gives (A.55). $\square$

A.9 Main Convergence Theorem

Proposition 2 (Average aggregation-error budget). *Under Assumptions 4, 5, 6, 7, and 8, together with the HRAC history-state update rules (A.12)–(A.14), the code-aligned global-scale warm-start, and $0 < s_{\min} \leq G_H$, the HRAC aggregation error satisfies, for every horizon $T \geq 1$,*

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|e_t\|^2 \leq C_{\text{agg}}, \quad (\text{A.56})$$

where

$$C_{\text{clip}} := 6\mathcal{C}_{\text{gc}}, \quad (\text{A.57})$$

$$C_{\text{res}} := 6\mathcal{C}_{\text{rc}}, \quad (\text{A.58})$$

$$C_{\text{byz}} := 3\varepsilon_{w,\infty}^2 G_H^2 + 3c_g^2 \beta_\infty^2 G_H^2, \quad (\text{A.59})$$

$$C_{\text{agg}} := C_{\text{clip}} + C_{\text{res}} + C_{\text{byz}}. \quad (\text{A.60})$$

Proof. For the honest part of the decomposition, Proposition 1 applies the clipped-moment lemma of Sadiev et al. [38, Lemma 5.1] once to the predictable global cap and once to the residual cap certified by the large- c scale-tracking condition. Thus, for each round,

$$\mathbb{E} \left[\sum_{i \in \mathcal{H}} w_i^t \left\| \tilde{\Delta}_i^t - m_i^t \right\|^2 \right] \leq 2\mathcal{C}_{\text{gc}} + 2\mathcal{C}_{\text{rc}}.$$

Using Lemma 10, this contributes

$$3\mathbb{E}\left[\sum_{i \in \mathcal{H}} w_i^t \left\|\tilde{\Delta}_i^t - m_i^t\right\|^2\right] \leq C_{\text{clip}} + C_{\text{res}}.$$

For every t , Assumptions 4 and 8 give

$$3\varepsilon_{w,t}^2 G_H^2 \leq 3\varepsilon_{w,\infty}^2 G_H^2,$$

and Corollary 1 together with Lemma 6 gives $\left\|\tilde{\Delta}_j^t\right\| \leq B_{\max} = c_g G_H$, hence

$$3\beta_t \sum_{j \in \mathcal{B}} w_j^t \left\|\tilde{\Delta}_j^t\right\|^2 \leq 3\beta_t (\beta_t c_g^2 G_H^2) \leq 3c_g^2 \beta_\infty^2 G_H^2.$$

Combining these two displays yields C_{byz} . Finally, apply Lemma 10, sum over t , and divide by T . $\square$

Theorem 1 (Convergence to an explicit HRAC constant neighbourhood). *Suppose Assumptions 1 to 8 hold, and HRAC uses the history-state update rules (A.12)–(A.14), the code-aligned global-scale warm-start, and a floor $0 < s_{\min} \leq G_H$. Let the step size be constant and satisfy*

$$0 < \eta \leq \frac{1}{L}. \quad (\text{A.61})$$

Define the explicit HRAC neighbourhood radius

$$R_{\text{HRAC}} := 2\xi^2 + 2C_{\text{agg}}. \quad (\text{A.62})$$

Then, for every $T \geq 1$,

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla f(x_t)\|^2 \leq \frac{2(f(x_0) - f^*)}{\eta T} + R_{\text{HRAC}}. \quad (\text{A.63})$$

Consequently,

$$\limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla f(x_t)\|^2 \leq R_{\text{HRAC}}. \quad (\text{A.64})$$

Therefore the averaged squared gradient norm converges to a first-order stationary neighbourhood with an explicit, time-independent radius.

Proof. Since $\eta \leq 1/L$, sum the sharpened one-step descent inequality (A.55) from $t = 0$ to $T - 1$:

$$\begin{aligned} \mathbb{E}[f(x_T)] &\leq f(x_0) - \frac{\eta}{2} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla f(x_t)\|^2 \\ &\quad + \eta \sum_{t=0}^{T-1} \mathbb{E} \|e_t\|^2 + \eta T \xi^2. \end{aligned}$$

Since $f(x_T) \geq f^*$, rearranging gives

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla f(x_t)\|^2 \leq \frac{2(f(x_0) - f^*)}{\eta T} + \frac{2}{T} \sum_{t=0}^{T-1} \mathbb{E} \|e_t\|^2 + 2\xi^2.$$

By Proposition 2,

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|e_t\|^2 \leq C_{\text{agg}}.$$

Substituting these average bounds yields (A.63). Taking lim sup eliminates the vanishing optimization term $2(f(x_0) - f^*)/(\eta T)$ and gives (A.64). $\square$

Corollary 2 (Closed-form HRAC constant interval). *Under the assumptions of Theorem 1,*

$$\begin{aligned} R_{\text{HRAC}} &= 2\xi^2 \\ &\quad + 12\mathcal{C}_{\text{gc}} + 12\mathcal{C}_{\text{rc}} + 6\varepsilon_{w,\infty}^2 G_H^2 + 6c_g^2 \beta_\infty^2 G_H^2. \end{aligned} \tag{A.65}$$

Hence the averaged stationarity measure is asymptotically confined to the explicit interval

$$0 \leq \limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla f(x_t)\|^2 \leq R_{\text{HRAC}}. \tag{A.66}$$

Remark 3 (Same theorem in schematic constant form). If one prefers the theorem-level presentation

$$R_{\text{HRAC}} = 2\xi^2 + C_{\text{clip}}^{\text{eff}} + C_{\text{res}}^{\text{eff}} + C_{\text{byz}}^{\text{eff}},$$

then the effective constants in the present proof are

$$C_{\text{clip}}^{\text{eff}} := 12\mathcal{C}_{\text{gc}}, \quad C_{\text{res}}^{\text{eff}} := 12\mathcal{C}_{\text{rc}}, \quad C_{\text{byz}}^{\text{eff}} := 6\varepsilon_{w,\infty}^2 G_H^2 + 6c_g^2 \beta_\infty^2 G_H^2.$$

The factor 2 relative to the raw aggregation-error constants (A.57)–(A.59) comes from the descent step $\frac{2}{T} \sum_t \mathbb{E} \|e_t\|^2$ in (A.63). When $\mathcal{C}_{\text{gc}}$ and $\mathcal{C}_{\text{rc}}$ are expanded using the clipped-moment constant 18 from Sadiev et al. [38, Lemma 5.1], the coefficient in front of the displayed moment terms becomes

$216 = 12 \times 18$. Here 18 is inherited from Sadiev et al. [38, Lemma 5.1], while $12 = 2 \times 3 \times 2$ comes respectively from the final descent conversion, the three-term aggregation-error decomposition, and the two-component clipping decomposition in Proposition 1.

Corollary 3 (Interpretation as an explicit stationary neighbourhood). *Under the assumptions of Theorem 1, HRAC converges sublinearly to a first-order stationary neighbourhood of radius R_{HRAC} . In particular, the long-run averaged squared gradient norm is controlled by a closed constant composed of:*

1. honest conditional-centre bias $2\xi^2$;
2. honest global-clipping constant $12\mathcal{C}_{\text{gc}}$;
3. residual Sadiev clipping constant $12\mathcal{C}_{\text{rc}}$;
4. residual Byzantine mass and honest-weight distortion $6\varepsilon_{w,\infty}^2 G_H^2 + 6c_g^2 \beta_\infty^2 G_H^2$.

A.10 Weight Control from a d -Gap Induced ν -Gap

The convergence theorem above only needs a closed upper bound on the Byzantine weight mass β_t . This section closes that bound through HRAC's adaptive statistics: persistent attack-induced self-inconsistency yields a positive drift in the residual-motion statistic d_i^t , the EMA recursion accumulates that drift into a visible late-stage ν -gap, and a realised/log-verifiable ν -gap certificate then yields explicit Byzantine weight suppression.

Notation for this section. $\bar{m}_{\mathcal{H}}^t = \frac{1}{H} \sum_{i \in \mathcal{H}} m_i^t$ is the honest conditional-centre field. Let a_i^t denote the client-level effective update centre tracked by HRAC's statistics path before residual clipping. For a Byzantine client j , write

$$a_j^t = \bar{m}_{\mathcal{H}}^t + \delta_{j,t},$$

where $\delta_{j,t}$ is simply the attack-induced displacement relative to the honest conditional-centre field. The vector q_i^t denotes the statistics-path clipped residual of client i , and $d_i^t := \|q_i^t - q_i^{t-1}\|$ is the per-round residual-motion statistic whose EMA is ν_i^t . The implemented excess score derived from ν_i^t is denoted by χ_i^t , and $\beta_t = \sum_{j \in \mathcal{B}} w_j^t$ is the total Byzantine aggregation weight.

A.10.1 Bounded-influence training dynamics

The optimization proof has already reduced HRAC to an honest training field plus a bounded aggregation perturbation:

$$g_t = \bar{m}_{\mathcal{H}}^t + e_t, \quad x_{t+1} = x_t - \eta_t(\bar{m}_{\mathcal{H}}^t + e_t). \quad (\text{A.67})$$

Here g_t is the actual HRAC aggregate used by the server, $\bar{m}_{\mathcal{H}}^t$ is the average honest conditional update centre, $e_t = g_t - \bar{m}_{\mathcal{H}}^t$ is the aggregation error bounded in Proposition 2, and η_t is the learning rate. Thus clipping is not used to claim that Byzantine updates disappear. Its role is to keep their per-round optimization influence finite while honest-majority training remains the dominant dynamical reference. This is aligned with the bounded-influence principle used by centered clipping methods for Byzantine optimization: Karimireddy et al. [6, Algorithm 1] clip each worker vector around a reference/history point, schematically

$$z_i = c + \text{clip}(v_i - c, \tau),$$

where v_i is worker i 's submitted vector, c is the current centre or history reference, τ is the clipping radius, and z_i is the message after centered clipping. Thus even if v_i is Byzantine, the deviation $z_i - c$ is bounded by τ .

The important point is that HRAC's history centre is client-specific. HRAC does not need a malicious client to be far from a global centre. Instead, it measures whether the client's residual sequence remains stable relative to its own EMA history. This is exactly the quantity computed by the code:

$$d_i^t = \|q_i^t - q_i^{t-1}\|,$$

where q_i^t is the statistics-path clipped residual for client i . The following elementary identity explains why a time-varying attack remains visible even when the malicious client has its own history centre. In an idealized deterministic notation, let

$$b_i^{t+1} = \rho_b b_i^t + (1 - \rho_b) a_i^t, \quad e_i^t := a_i^t - b_i^t.$$

Then

$$e_i^{t+1} = \rho_b e_i^t + (a_i^{t+1} - a_i^t). \quad (\text{A.68})$$

Thus the history residual is a stable high-pass trace of the client's own update increments. For an honest client, $a_i^{t+1} - a_i^t$ is mainly ordinary training drift plus stochastic noise. For a Byzantine client $a_j^t = \bar{m}_{\mathcal{H}}^t + \delta_{j,t}$, the increment contains the additional attack-field increment

$$\delta_{j,t+1} - \delta_{j,t}.$$

The EMA centre can absorb slowly varying behaviour, but it lags behind a persistent time-varying attack displacement. After global and residual clipping, the amplitude of this displacement is bounded, but the remaining temporal self-inconsistency is precisely what contributes to $\|q_j^t - q_j^{t-1}\|$.

A.10.2 A literature-supported route to persistent residual-motion drift

The source of the ν -gap is not a large raw norm. It is a persistent temporal mismatch between a client's current effective update and the behaviour that its own EMA history has already absorbed. Two classic attacks make this mechanism plausible. First, Xie et al. [39] construct an inner-product manipulation (IPM) attack whose Krum version sets all Byzantine gradients to

$$u_1 = \dots = u_q = -\epsilon \bar{v}, \quad \bar{v} = \frac{1}{m-q} \sum_{i=1}^{m-q} v_i,$$

where q is the number of Byzantine workers, $m-q$ is the number of correct workers, v_i are the correct gradients, $\bar{v}$ is their mean, and $\epsilon > 0$ controls the attack magnitude in the opposite direction. Identifying $\bar{v}$ with the honest centre $\bar{m}_{\mathcal{H}}^t$ gives, in the notation above,

$$a_j^t = -\epsilon \bar{m}_{\mathcal{H}}^t, \quad \delta_{j,t}^{\text{IPM}} = -(1+\epsilon) \bar{m}_{\mathcal{H}}^t.$$

Thus IPM contributes the attack-field increment

$$\delta_{j,t}^{\text{IPM}} - \delta_{j,t-1}^{\text{IPM}} = -(1+\epsilon)(\bar{m}_{\mathcal{H}}^t - \bar{m}_{\mathcal{H}}^{t-1}).$$

When the honest training field changes along the trajectory (A.67), the malicious history must track a time-varying negative multiple of that field. In such rounds, the unabsorbed part can appear in $\|q_j^t - q_j^{t-1}\|$.

Second, Baruch et al. [8] propose the ALIE coordinate-wise perturbation

$$(p_{\text{mal}})_k = \mu_k - z_{\text{max}}\sigma_k,$$

where μ_k and σ_k are coordinate mean and standard-deviation estimates, z_{max} is the attack's allowed standard-deviation multiplier, and $(p_{\text{mal}})_k$ is the malicious value submitted on coordinate k . Writing these round- t coordinate estimates as $\mu_{t,k}^{\text{coord}}$ and $\sigma_{t,k}^{\text{coord}}$, and identifying $\mu_{t,k}^{\text{coord}}$ with the k -th coordinate of $\bar{m}_{\mathcal{H}}^t$, the attack displacement is

$$(\delta_{j,t}^{\text{ALIE}})_k \approx -z_{\text{max}}\sigma_{t,k}^{\text{coord}}.$$

Consequently,

$$(\delta_{j,t}^{\text{ALIE}} - \delta_{j,t-1}^{\text{ALIE}})_k \approx -z_{\text{max}}(\sigma_{t,k}^{\text{coord}} - \sigma_{t-1,k}^{\text{coord}}).$$

Thus changing coordinate scales can induce diagonal attack-field increments without requiring large update norms. HRAC clipping bounds their amplitude, while the client-specific EMA history absorbs them only with lag. When this attack field varies along the training trajectory, the unabsorbed part can appear in the same full-vector residual-motion statistic that the code uses.

Let q_i^t denote the statistics-path clipped residual used by HRAC, and define

$$d_i^t := \|q_i^t - q_i^{t-1}\|. \quad (\text{A.69})$$

Here q_i^t is the statistics-path clipped residual for client i , so d_i^t measures the full-vector change in that client's residual behaviour from round $t-1$ to round t . For group means, write

$$\bar{d}_H^t := \frac{1}{H} \sum_{i \in \mathcal{H}} d_i^t, \quad \bar{d}_B^t := \frac{1}{B} \sum_{j \in \mathcal{B}} d_j^t. \quad (\text{A.70})$$

Thus $\bar{d}_H^t$ and $\bar{d}_B^t$ are the honest and Byzantine group averages of the residual-motion statistic.

Condition 1 (Persistent group d -drift). *There exist $T_d < \infty$ and $\gamma_d > 0$ such that, for all $t \geq T_d$,*

$$\mathbb{E}[\bar{d}_B^t - \bar{d}_H^t \mid \mathcal{F}_{t-1}] \geq \gamma_d. \quad (\text{A.71})$$

Writing $D_t := \bar{d}_B^t - \bar{d}_H^t$, this is a persistent conditional drift condition. It does not require every realised round to have a visible positive gap. Equivalently, one may think of $D_t = \gamma_d + \varepsilon_t$, where the noise term ε_t can hide the gap early in training, while $\mathbb{E}[\varepsilon_t \mid \mathcal{F}_{t-1}] \geq 0$ preserves the positive

drift that the ν -EMA accumulates over time.

Proposition 3 (Persistent d -drift is accumulated into a ν -gap). *Suppose $0 < \rho_\nu < 1$ and*

$$\nu_i^{t+1} = \rho_\nu \nu_i^t + (1 - \rho_\nu) d_i^t. \quad (\text{A.72})$$

If Condition 1 holds, then with

$$z_t := \bar{\nu}_B^t - \bar{\nu}_H^t,$$

we have, for all $t \geq T_d$,

$$\mathbb{E}[z_{t+1}] \geq \rho_\nu \mathbb{E}[z_t] + (1 - \rho_\nu) \gamma_d. \quad (\text{A.73})$$

Consequently, for every $t \geq T_d$,

$$\mathbb{E}[z_t] \geq \rho_\nu^{t-T_d} \mathbb{E}[z_{T_d}] + (1 - \rho_\nu^{t-T_d}) \gamma_d, \quad (\text{A.74})$$

and therefore

$$\liminf_{t \rightarrow \infty} \mathbb{E}[\bar{\nu}_B^t - \bar{\nu}_H^t] \geq \gamma_d. \quad (\text{A.75})$$

Proof. Averaging (A.72) over Byzantine and honest clients and subtracting gives

$$z_{t+1} = \rho_\nu z_t + (1 - \rho_\nu)(\bar{d}_B^t - \bar{d}_H^t).$$

Taking expectation and applying Condition 1 yields (A.73). Unrolling this affine recursion gives (A.74), and taking $\liminf$ gives (A.75). $\square$

Thus a literature-supported persistent residual self-inconsistency is formalized directly as the group d -drift Condition 1, using exactly the full-vector statistic computed by HRAC. Proposition 3 then shows that HRAC's EMA statistic accumulates this persistent d -drift into a late-stage ν -gap. This matches the empirical pattern: the underlying d -drift can be present before it is visually obvious, while early stochastic noise and history warm-up obscure the realised separation. The EMA recursion acts as a low-pass accumulator, so a small persistent drift becomes clearer in ν after enough rounds.

Remark 4 (Expectation-level route versus realised weight control). Proposition 3 gives an expectation-level origin for the group ν -gap. The implemented weights, however, are computed from the realised statistics, not only from their expectations. The weight-control argument below therefore uses a late-stage realised ν -gap certificate. The expectation-level route can be upgraded to this certificate

under an additional concentration bound or an empirical log certificate showing that the realised group means and radii stay close to their expectations on the late-stage window.

When this appendix is read together with the dissertation experiments, the diagnostic plots provide qualitative realised-trajectory evidence of this kind. They are not used as a formal numerical certificate in the theorem. In particular, Figure 4.9, Figure 4.13, and Figure 4.14 report the logged HRAC residual-motion statistics and weights under label flipping, ALIE, and IPM, respectively. These plots are used only as realised-trajectory evidence for Condition 2: on a selected late-stage window, one can read or compute the group means $\bar{\nu}_H^t, \bar{\nu}_B^t$ and radii r_H^t, r_B^t , then certify concrete values of $\bar{\gamma}_\nu, r_H, r_B$.

Definition 1 (Late-stage ν group means and radii). *For each round t , define*

$$\bar{\nu}_H^t := \frac{1}{H} \sum_{i \in \mathcal{H}} \nu_i^t, \quad \bar{\nu}_B^t := \frac{1}{B} \sum_{j \in \mathcal{B}} \nu_j^t.$$

The within-group ν -radii are

$$r_H^t := \max_{i \in \mathcal{H}} |\nu_i^t - \bar{\nu}_H^t|, \quad r_B^t := \max_{j \in \mathcal{B}} |\nu_j^t - \bar{\nu}_B^t|.$$

Condition 2 (Late-stage realised ν -gap certificate). *There exist $T_\nu < \infty$, $\bar{\gamma}_\nu > 0$, and radii $r_H, r_B \geq 0$ such that for all $t \geq T_\nu$,*

$$\bar{\nu}_B^t - \bar{\nu}_H^t \geq \bar{\gamma}_\nu, \quad r_H^t \leq r_H, \quad r_B^t \leq r_B. \quad (\text{A.76})$$

Remark 5 (Logged closing certificate). Condition 2 is not an additional algorithmic mechanism. It is a realised-trajectory closing certificate: on a late-stage window of the experimental logs, one reads or computes $\bar{\nu}_H^t, \bar{\nu}_B^t, r_H^t, r_B^t$ and checks that the displayed inequalities hold with explicit constants. Once certified, these constants close the remaining gap between the expectation-level d -drift argument and the pathwise weight bound used by the implemented HRAC rule.

Proposition 4 (Realised mean ν -gap gives an implemented score gap). *Suppose the realised late-stage logs certify Condition 2. Define*

$$\bar{\nu}_t := \frac{1}{N} \sum_{k=1}^N \nu_k^t = \frac{H\bar{\nu}_H^t + B\bar{\nu}_B^t}{N}, \quad \chi_i^t := (\nu_i^t - \bar{\nu}_t)_+.$$

Then for all $t \geq T_\nu$,

$$\min_{j \in \mathcal{B}} \chi_j^t \geq \chi_B^\nu := \left(\frac{H}{N} \bar{\gamma}_\nu - r_B \right)_+, \quad (\text{A.77})$$

$$\max_{i \in \mathcal{H}} \chi_i^t \leq \chi_H^\nu := \left(r_H - \frac{B}{N} \bar{\gamma}_\nu \right)_+. \quad (\text{A.78})$$

Proof. For $j \in \mathcal{B}$,

$$\nu_j^t - \bar{\nu}_t \geq \bar{\nu}_B^t - r_B - \frac{H\bar{\nu}_H^t + B\bar{\nu}_B^t}{N} = \frac{H}{N}(\bar{\nu}_B^t - \bar{\nu}_H^t) - r_B \geq \frac{H}{N} \bar{\gamma}_\nu - r_B.$$

Taking positive parts gives (A.77). Similarly, for $i \in \mathcal{H}$,

$$\nu_i^t - \bar{\nu}_t \leq \bar{\nu}_H^t + r_H - \frac{H\bar{\nu}_H^t + B\bar{\nu}_B^t}{N} = r_H - \frac{B}{N}(\bar{\nu}_B^t - \bar{\nu}_H^t) \leq r_H - \frac{B}{N} \bar{\gamma}_\nu.$$

Taking positive parts gives (A.78). □

Corollary 4 (Closed Byzantine-mass constant from the realised ν -gap). *This corollary is an idealised late-stage weight-control model. It is not a line-by-line proof of the implemented weighting rule, which also includes the $\nu_i > 0.9$ activation threshold, EMA smoothing of the penalty score, and an iterative maximum-weight cap. Suppose the realised late-stage logs certify Condition 2, so that the score bounds of Proposition 4 hold. Suppose that, after a finite time $T_\star$, the late-stage weights are approximated by the inactive-cap inverse-score form*

$$\hat{w}_i^t = \frac{1}{1 + \alpha_w \chi_i^t}, \quad w_i^t = \frac{\hat{w}_i^t}{\sum_{k=1}^N \hat{w}_k^t},$$

where $\alpha_w > 0$ is the penalty strength, and suppose $\chi_B^\nu > \chi_H^\nu$. Define

$$T_{\text{close}} := \max\{T_\star, T_\nu\}.$$

Then for all $t \geq T_{\text{close}}$,

$$\beta_t := \sum_{j \in \mathcal{B}} w_j^t \leq \beta_\infty^\nu := \frac{B\chi_B^\nu}{Hu_H^\nu + B\chi_B^\nu}, \quad u_B^\nu := \frac{1}{1 + \alpha_w \chi_B^\nu}, \quad u_H^\nu := \frac{1}{1 + \alpha_w \chi_H^\nu}. \quad (\text{A.79})$$

If, in addition, for all $t \geq T_{\text{close}}$,

$$\varepsilon_{w,t} \leq \varepsilon_{w,\infty}^\nu,$$

then the Byzantine part of the asymptotic HRAC radius is controlled by

$$C_{\text{byz}}^\nu := 3(\varepsilon_{w,\infty}^\nu)^2 G_H^2 + 3c_g^2 (\beta_\infty^\nu)^2 G_H^2. \quad (\text{A.80})$$

Proof. For Byzantine clients, Proposition 4 gives $\chi_j^t \geq \chi_B^\nu$, hence $\hat{w}_j^t \leq u_B^\nu$. For honest clients, it gives $\chi_i^t \leq \chi_H^\nu$, hence $\hat{w}_i^t \geq u_H^\nu$. Therefore

$$\beta_t = \frac{\sum_{j \in \mathcal{B}} \hat{w}_j^t}{\sum_{i \in \mathcal{H}} \hat{w}_i^t + \sum_{j \in \mathcal{B}} \hat{w}_j^t} \leq \frac{Bu_B^\nu}{Hu_H^\nu + Bu_B^\nu}.$$

The closed Byzantine constant follows by substituting this bound into the weight-distortion and Byzantine-energy terms of Lemma 10; the Byzantine energy part uses the pathwise bound $\|\tilde{\Delta}_j^t\| \leq B_{\max} = c_g G_H$ from Lemma 6. $\square$

Corollary 5 (Main HRAC interval closed by Sadiev clipping and the realised ν -gap certificate). *Suppose the assumptions of Theorem 1 hold, except that the constant β_∞ in Assumption 8 is only required after a finite transient and is supplied by Corollary 4. Suppose also that the per-round clipping controls are certified by Proposition 1: the global control by the clipped-moment lemma of Sadiev et al. [38, Lemma 5.1] applied to the predictable history-EMA cap, and the residual control by the same lemma applied under the large- c residual scale-tracking condition. Thus*

$$\mathcal{C}_{\text{gc}} = 18B_{\max}^{2-\alpha_g} \sigma_{\text{gc}}^{\alpha_g}, \quad \mathcal{C}_{\text{rc}} = 18\tau_{\max}^{2-\alpha_r} \sigma_{\text{rc}}^{\alpha_r}.$$

The coefficients 216 in the next display are exactly 12×18 : the factor 18 is the second-moment constant in Sadiev et al. [38, Lemma 5.1], and the factor 12 is the proof-level amplification explained in Remark 3. Then

$$\begin{aligned} \limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla f(x_t)\|^2 &\leq 2\xi^2 \\ &\quad + 216B_{\max}^{2-\alpha_g} \sigma_{\text{gc}}^{\alpha_g} + 216\tau_{\max}^{2-\alpha_r} \sigma_{\text{rc}}^{\alpha_r} + 6(\varepsilon_{w,\infty}^\nu)^2 G_H^2 + 6c_g^2 (\beta_\infty^\nu)^2 G_H^2. \end{aligned} \quad (\text{A.81})$$

Proof. Let $T_{\text{close}} := \max\{T_\star, T_\nu\}$. Apply the main descent proof from T_{close} onward. The finite prefix $t < T_{\text{close}}$ contributes only $O(1/T)$ to the averaged stationarity measure: the prefix length is finite, and the deterministic cap chain bounds g_t , $\bar{m}_{\mathcal{H}}^t$, and e_t , so its total contribution is a finite constant divided by T . Proposition 1 supplies the two per-round clipping constants: the global one through the clipped-moment lemma of Sadiev et al. [38, Lemma 5.1] applied to $(X, x, \lambda) =$

(Δ_i^t, m_i^t, B_t) , and the residual one through the same lemma applied to $(X, x, \lambda) = (r_i^t, s_i^t, \tau_i^t)$, with admissibility supplied by (A.24). Corollary 4 supplies the closed Byzantine mass constant from the realised late-stage ν -gap certificate. Taking $\limsup$ gives (A.81). $\square$

Bibliography

- [1] H. B. McMahan, E. Moore, D. Ramage, and B. A. y Arcas, “Federated learning of deep networks using model averaging,” *CoRR*, vol. abs/1602.05629, 2016. arXiv: 1602.05629. [Online]. Available: <http://arxiv.org/abs/1602.05629>
- [2] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y. Arcas, “Communication-efficient learning of deep networks from decentralized data,” in *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*, A. Singh and J. Zhu, Eds., ser. Proceedings of Machine Learning Research, vol. 54, PMLR, Apr. 2017, pp. 1273–1282. [Online]. Available: <https://proceedings.mlr.press/v54/mcmahan17a.html>
- [3] P. Kairouz et al., “Advances and open problems in federated learning,” *CoRR*, vol. abs/1912.04977, 2019. arXiv: 1912.04977. [Online]. Available: <http://arxiv.org/abs/1912.04977>
- [4] L. Lamport, R. Shostak, and M. Pease, “The byzantine generals problem,” *ACM Trans. Program. Lang. Syst.*, vol. 4, no. 3, pp. 382–401, Jul. 1982, issn: 0164-0925. DOI: 10.1145/357172.357176 [Online]. Available: <https://doi.org/10.1145/357172.357176>
- [5] P. Blanchard, E. M. El Mhamdi, R. Guerraoui, and J. Stainer, “Machine learning with adversaries: Byzantine tolerant gradient descent,” in *Advances in Neural Information Processing Systems*, I. Guyon et al., Eds., vol. 30, Curran Associates, Inc., 2017. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2017/file/f4b9ec30ad9f68f89b29639786cb62ef-Paper.pdf
- [6] S. P. Karimireddy, L. He, and M. Jaggi, “Learning from history for byzantine robust optimization,” in *Proceedings of the 38th International Conference on Machine Learning*, M. Meila and T. Zhang, Eds., ser. Proceedings of Machine Learning Research, vol. 139, PMLR, Jul. 2021, pp. 5311–5319. [Online]. Available: <https://proceedings.mlr.press/v139/karimireddy21a.html>
- [7] J. Peng, W. Li, S. Vlaski, and Q. Ling, “Mean aggregator is more robust than robust aggregators under label poisoning attacks on distributed heterogeneous data,” *Journal of Machine Learning Research*, vol. 26, no. 27, pp. 1–51, 2025. [Online]. Available: <http://jmlr.org/papers/v26/24-1307.html>

-
- [8] M. Baruch, G. Baruch, and Y. Goldberg, *A little is enough: Circumventing defenses for distributed learning*, 2019. arXiv: 1902.06156 [cs.LG]. [Online]. Available: <https://arxiv.org/abs/1902.06156>
 - [9] K. Bonawitz et al., “Towards federated learning at scale: System design,” in *Proceedings of Machine Learning and Systems*, vol. 1, 2019, pp. 374–388. [Online]. Available: https://proceedings.mlsys.org/paper_files/paper/2019/hash/7b770da633baf74895be22a8807f1a8f-Abstract.html
 - [10] J. Wang et al., “A field guide to federated optimization,” *CoRR*, vol. abs/2107.06917, 2021. arXiv: 2107.06917. [Online]. Available: <https://arxiv.org/abs/2107.06917>
 - [11] S. Caldas et al., “LEAF: A benchmark for federated settings,” *CoRR*, vol. abs/1812.01097, 2018. arXiv: 1812.01097. [Online]. Available: <https://arxiv.org/abs/1812.01097>
 - [12] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, “Federated optimization in heterogeneous networks,” in *Proceedings of Machine Learning and Systems*, vol. 2, 2020, pp. 429–450. [Online]. Available: https://proceedings.mlsys.org/paper_files/paper/2020/hash/1f5fe83998a09396ebe6477d9475ba0c-Abstract.html
 - [13] S. P. Karimireddy, S. Kale, M. Mohri, S. J. Reddi, S. U. Stich, and A. T. Suresh, “SCAF-FOLD: stochastic controlled averaging for on-device federated learning,” *CoRR*, vol. abs/1910.06378, 2019. arXiv: 1910.06378. [Online]. Available: <http://arxiv.org/abs/1910.06378>
 - [14] S. J. Reddi et al., “Adaptive federated optimization,” in *International Conference on Learning Representations*, 2021. [Online]. Available: <https://openreview.net/forum?id=LkFG31B13U5>
 - [15] T.-M. H. Hsu, H. Qi, and M. Brown, “Measuring the effects of non-identical data distribution for federated visual classification,” *CoRR*, vol. abs/1909.06335, 2019. arXiv: 1909.06335. [Online]. Available: <https://arxiv.org/abs/1909.06335>
 - [16] L. He, S. P. Karimireddy, and M. Jaggi, “Byzantine-robust learning on heterogeneous datasets via resampling,” *CoRR*, vol. abs/2006.09365, 2020. arXiv: 2006.09365. [Online]. Available: <https://arxiv.org/abs/2006.09365>
 - [17] Y. Allouah, S. Farhadkhani, R. Guerraoui, N. Gupta, R. Pinot, and J. Stephan, “Fixing by mixing: A recipe for optimal byzantine ml under heterogeneity,” in *Proceedings of The 26th International Conference on Artificial Intelligence and Statistics*, F. Ruiz, J. Dy, and J.-W. van de Meent, Eds., ser. Proceedings of Machine Learning Research, vol. 206, PMLR,

- Apr. 2023, pp. 1232–1300. [Online]. Available: <https://proceedings.mlr.press/v206/allouah23a.html>
- [18] R. Guerraoui, N. Gupta, and R. Pinot, “Byzantine machine learning: A primer,” *ACM Comput. Surv.*, vol. 56, no. 7, Apr. 2024, ISSN: 0360-0300. DOI: 10.1145/3616537 [Online]. Available: <https://doi.org/10.1145/3616537>
- [19] Q. Xia, Z. Tao, Z. Hao, and Q. Li, “FABA: An algorithm for fast aggregation against byzantine attacks in distributed neural networks,” in *Proceedings of the Twenty-Eighth International Joint Conference on Artificial Intelligence*, 2019, pp. 4824–4830. DOI: 10.24963/ijcai.2019/670 [Online]. Available: <https://www.ijcai.org/proceedings/2019/670>
- [20] D. Yin, Y. Chen, R. Kannan, and P. Bartlett, “Byzantine-robust distributed learning: Towards optimal statistical rates,” in *Proceedings of the 35th International Conference on Machine Learning*, J. Dy and A. Krause, Eds., ser. Proceedings of Machine Learning Research, vol. 80, PMLR, Jul. 2018, pp. 5650–5659. [Online]. Available: <https://proceedings.mlr.press/v80/yin18a.html>
- [21] K. Pillutla, S. M. Kakade, and Z. Harchaoui, “Robust aggregation for federated learning,” *IEEE Transactions on Signal Processing*, vol. 70, pp. 1142–1154, 2022, ISSN: 1941-0476. DOI: 10.1109/tsp.2022.3153135 [Online]. Available: <http://dx.doi.org/10.1109/TSP.2022.3153135>
- [22] L. Li, W. Xu, T. Chen, G. B. Giannakis, and Q. Ling, “RSA: Byzantine-robust stochastic aggregation methods for distributed learning from heterogeneous datasets,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 33, 2019, pp. 1544–1551. DOI: 10.1609/aaai.v33i01.33011544 [Online]. Available: <https://doi.org/10.1609/aaai.v33i01.33011544>
- [23] O. Thakkar, G. Andrew, and H. B. McMahan, “Differentially private learning with adaptive clipping,” *CoRR*, vol. abs/1905.03871, 2019. arXiv: 1905.03871. [Online]. Available: <http://arxiv.org/abs/1905.03871>
- [24] E. M. El Mhamdi, R. Guerraoui, and S. Rouault, “The hidden vulnerability of distributed learning in Byzantium,” in *Proceedings of the 35th International Conference on Machine Learning*, J. Dy and A. Krause, Eds., ser. Proceedings of Machine Learning Research, vol. 80, PMLR, Jul. 2018, pp. 3521–3530. [Online]. Available: <https://proceedings.mlr.press/v80/mhamdi18a.html>

-
- [25] S. Vlaski, C. Schroth, M. Muma, and A. M. Zoubir, “Robust and efficient aggregation for distributed learning,” in *2022 30th European Signal Processing Conference (EUSIPCO)*, 2022, pp. 817–821. DOI: 10.23919/EUSIPCO55093.2022.9909822
- [26] M. Fang, X. Cao, J. Jia, and N. Z. Gong, “Local model poisoning attacks to byzantine-robust federated learning,” in *Proceedings of the 29th USENIX Conference on Security Symposium*, ser. SEC’20, USA: USENIX Association, 2020, ISBN: 978-1-939133-17-5.
- [27] V. Shejwalkar, A. Houmansadr, P. Kairouz, and D. Ramage, “Back to the drawing board: A critical evaluation of poisoning attacks on production federated learning,” in *2022 IEEE Symposium on Security and Privacy (SP)*, 2022, pp. 1354–1371. DOI: 10.1109/SP46214.2022.9833647
- [28] M. Yang, H. Cheng, F. Chen, X. Liu, M. Wang, and X. Li, “Model poisoning attack in differential privacy-based federated learning,” *Information Sciences*, vol. 630, pp. 158–172, 2023, ISSN: 0020-0255. DOI: <https://doi.org/10.1016/j.ins.2023.02.025> [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0020025523002141>
- [29] V. Tolpegin, S. Truex, M. E. Gursoy, and L. Liu, “Data poisoning attacks against federated learning systems,” in *Computer Security – ESORICS 2020*, L. Chen, N. Li, K. Liang, and S. Schneider, Eds., Cham: Springer International Publishing, 2020, pp. 480–501, ISBN: 978-3-030-58951-6.
- [30] P. Tavallali, V. Behzadan, A. Alizadeh, A. Ranganath, and M. Singhal, “Adversarial label-poisoning attacks and defense for general multi-class models based on synthetic reduced nearest neighbor,” in *2022 IEEE International Conference on Image Processing (ICIP)*, 2022, pp. 3717–3722. DOI: 10.1109/ICIP46576.2022.9897807
- [31] N. M. Jebreel, J. Domingo-Ferrer, D. Sánchez, and A. Blanco-Justicia, “Lfighter: Defending against the label-flipping attack in federated learning,” *Neural Networks*, vol. 170, pp. 111–126, 2024, ISSN: 0893-6080. DOI: <https://doi.org/10.1016/j.neunet.2023.11.019> [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0893608023006421>
- [32] Z. Wu, Q. Ling, T. Chen, and G. B. Giannakis, “Federated variance-reduced stochastic gradient descent with robustness to byzantine attacks,” *IEEE Transactions on Signal Processing*, vol. 68, pp. 4583–4596, 2020. DOI: 10.1109/TSP.2020.3012952 [Online]. Available: <https://doi.org/10.1109/TSP.2020.3012952>

-
- [33] S. Farhadkhani, R. Guerraoui, N. Gupta, R. Pinot, and J. Stephan, “Byzantine machine learning made easy by resilient averaging of momentums,” in *Proceedings of the 39th International Conference on Machine Learning*, K. Chaudhuri, S. Jegelka, L. Song, C. Szepesvari, G. Niu, and S. Sabato, Eds., ser. Proceedings of Machine Learning Research, vol. 162, PMLR, 2022, pp. 6246–6283. [Online]. Available: <https://proceedings.mlr.press/v162/farhadkhani22a.html>
- [34] J. H. Choi, Y. Liu, and Y. Maurya, *Is it worth storing historical gradients?* Course project for Federated and Collaborative Learning, Carnegie Mellon University, 2024. [Online]. Available: <https://yashmaurya.com/pdfs/GradientsFL.pdf>
- [35] S. Zhang, Z. Jiang, H. Gao, S. Gui, and T. Gao, “Attacks and countermeasures on federated learning via historical knowledge modeling,” *Journal of King Saud University – Computer and Information Sciences*, vol. 37, p. 116, 2025, Published online: 17 July 2025. DOI: 10.1007/s44443-025-00115-1 [Online]. Available: <https://doi.org/10.1007/s44443-025-00115-1>
- [36] K. Bonawitz et al., “Practical secure aggregation for privacy-preserving machine learning,” in *Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security*, ser. CCS ’17, Dallas, Texas, USA: Association for Computing Machinery, 2017, pp. 1175–1191, ISBN: 9781450349468. DOI: 10.1145/3133956.3133982 [Online]. Available: <https://doi.org/10.1145/3133956.3133982>
- [37] X. Cao, M. Fang, J. Liu, and N. Z. Gong, “Fltrust: Byzantine-robust federated learning via trust bootstrapping,” *CoRR*, vol. abs/2012.13995, 2020. arXiv: 2012.13995. [Online]. Available: <https://arxiv.org/abs/2012.13995>
- [38] A. Sadiev et al., “High-probability bounds for stochastic optimization and variational inequalities: The case of unbounded variance,” in *Proceedings of the 40th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 202, PMLR, 2023, pp. 29 563–29 648. [Online]. Available: <https://proceedings.mlr.press/v202/sadiev23a.html>
- [39] C. Xie, O. Koyejo, and I. Gupta, “Fall of empires: Breaking byzantine-tolerant SGD by inner product manipulation,” in *Proceedings of the Thirty-Fifth Conference on Uncertainty in Artificial Intelligence*, ser. Proceedings of Machine Learning Research, vol. 115, PMLR, 2020, pp. 261–270. [Online]. Available: <https://proceedings.mlr.press/v115/xie20a.html>